



Affirmed Networks RADIUS Interface Specification

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- Message Definitions
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- RADIUS AVP Descriptions

Affirmed Networks Inc.

35 Nagog Park
Acton, MA 01720
USA

Tel: +1-978-268-0800

www.affirmednetworks.com

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Preface

This specification contains information on the RADIUS interface used for carrying authentication, authorization, and accounting (AAA) information between the Affirmed Networks Mobile Content Cloud (MCC) device and a centralized AAA (or RADIUS) server.

***Note:** The Affirmed Networks Mobile Content Cloud™ is the mobile services software solution that operates on virtual platforms such as blade systems.*

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide) working with RADIUS interfaces. The audience should be familiar with AAA services, the RADIUS protocol, and 3G or 4G networks.

Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Use This Guide

Read the following chapters and appendices in the order in which they appear:

Chapter/Appendix	Describes
Chapter 1, RADIUS Interface	The message flows, configuration steps, failover handling, and message definitions used to control and monitor RADIUS interfaces.
Chapter 2, RADIUS AVP Descriptions	The attribute value pair definitions for RADIUS interfaces.

References and Specifications

The system supports the following references and specifications for the RADIUS Authorization and Accounting interfaces:

- RFC 2865: *Remote Authentication Dial In User Service (RADIUS)*
- RFC 2866: *RADIUS Accounting*
- RFC 2867: *RADIUS Tunnel Accounting Support*
- RFC 2868: *RADIUS Attributes for Tunnel Protocol Support*
- RFC 2869: *RADIUS Extensions*
- RFC 3162: *RADIUS and IPv6*
- RFC 4372: *Chargeable User Identity*
- RFC 5176: *Dynamic Authorization Extensions to Remote Authentication Dial In User Service (RADIUS)*
- 3GPP TS 29.061 V11.0.0 (2012-03): *Interworking between the Public Land Mobile Network (PLMN) supporting packet based services and Packet Data Networks (PDN)*

Features in this Release

This release includes the following new features. Note the following:

- For a complete list of the features supported in this release, see the *Software Release Notes*.
- For a list of additional Affirmed Networks documentation, see the *Affirmed Networks Documentation Library*.
- For a comprehensive list of terms and acronyms used in Affirmed Networks documentation, see *Affirmed Networks Terms and Acronyms*.

Feature	Description
Added support for the TMO-Virtual-Gi-ID VSA on RADIUS, Gx, and Gy Interfaces See Processing the TMO-Virtual-Gi-ID VSA (page 1-56) .	This feature adds support for the TMO-Virtual-Gi-ID vendor-specific attribute (VSA) on RADIUS, Gx, and Gy interfaces. This feature also adds two new configuration parameters under the APN object to specify whether to process the TMO-Virtual-Gi-ID VSA when received in the Access-Accept message from the RADIUS server. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name> vsa-virtual-gi-id-enabled [true false] vsa-virtual-gi-id <default-value></pre>
Added support for new TMO-Transparent Data attribute for the RADIUS Access-Accept message and for the Diameter AVP on the Gy interface, and added support for a new CDR field on the Ga interface See Access-Accept (page 1-81) .	Added support for the following: ■ Transparent data AVP in CCR messages. This AVP is configurable in Gy CCR data record templates and by default is set to no, indicating the field is not sent. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ocs diameter [event interim start stop] <template-name> tmotransparentdata presentFlag [yes no]</pre> ■ Transparent data attribute in RADIUS Access Accept Message from the RADIUS Authentication Server. This attribute is configurable in RADIUS authorization accept data record templates and by default is set to no, indicating the field is not processed. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw auth radius access-accept <template-name> tmo_transparent_data presentFlag [yes no]</pre> ■ Transparent data subfield in the RecordExtensions field of GTPP CDRs (sent via the Ga interface). This subfield, transparentVSA, is configurable in GTPP data record templates and by default is set to no, indicating the field is not present in CDRs. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template [pgw sgw] ofcs gtp [interim stop] <template-name> extensioninfo presentFlag [yes no]</pre> Three other subfields have also been added to support the transparentVSA subfield: extensionType, length, and serviceList. To view the transparent data string for a subscriber on a PGW or SGW: <pre>an3000-mcm-slot7cpu1# show subscriber pdn-session <id></pre>

Feature	Description
Support for PGW selection of serving network MCC/MNC and TMO-Transparent-Data See the <i>Affirmed Networks Gateway Administration Guide</i> .	This feature lets the PGW select MCC/MNC (PLMN ID) values in a configured priority order by using the GTP message from the SGSN/SGW, a sig-info-table, or the RADIUS Access-Accept Message from the RADIUS Authentication Server. This feature also lets you configure the selection of TMO-Transparent Data values on the PGW when the RADIUS Access-Accept Message does not return a transparent data value. The PGW selects a value from a sig-info-table with PLMN ID mappings or from a sig-info-table with SGSN/SGW subnet mappings. This is enabled at the APN level.

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menus	Times New Roman bold font	Use configure_ha to configure... Enter y . To continue, press Enter .
Selecting a menu item	Menu => Option	Select Security => Group Management .
Variable parameters for which you supply values	<i><courier bold italic blue font></i>	configure_dr convert slave --masterhost <host IP>
User input	Courier bold blue font	cd /opt/Affirmed/NMS/bin
Screen messages, system output, sample source code	Courier regular blue font	Checking for License...
Required keywords and arguments in square brackets	[]	cluster [id] node [id] reboot
Optional keywords and arguments in curly brackets	{ }	{cdr-file-name}
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.		Is this correct? [y n]
Keywords representing a single element in angle brackets	< >	<i><host name></i>
Asterisk after a CLI object indicates a required object.	*	name*
Variables, book and chapter titles, file path names, new terms, and emphasized text	<i>Times New Roman italic font</i>	See the <i>Acuitas User's Guide</i> . <i>Workflow</i> is an application running on the system.

Tip: Helpful suggestions.

Note: *Additional information that may apply to the subject text.*



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RADIUS Interface

This section contains the following information used to control and monitor RADIUS interfaces:

- [Interface Overview \(page 1-2\)](#)
- [RADIUS Configuration \(page 1-8\)](#)
- [Optional RADIUS Configuration Tasks \(page 1-25\)](#)
- [Monitoring \(page 1-57\)](#)
- [Failover Handling \(page 1-61\)](#)
- [Message Definitions \(page 1-76\)](#)

See [Chapter 2, RADIUS AVP Descriptions](#) for a description of the AVPs supported on RADIUS interfaces.

Note: See the *Affirmed Networks CLI Reference Guide* for more information on the CLI commands used to configure and monitor RADIUS interfaces.

Interface Overview

Remote Authentication Dial-In User Service (RADIUS) is a client/server protocol used for carrying authentication, authorization, and accounting (AAA) information between Network Access Servers (NAS) and a centralized AAA (or RADIUS) server.

- *Authentication* – A process used to verify a user's identity before granting access to the requested services.
- *Authorization* – A process used to control the network services granted to a particular user after authentication.
- *Accounting* – A process used to generate log files based on user session information. These log files are used for billing and analyzing network usage.

The centralized RADIUS server authenticates users and authorizes network services based on the user profile information contained in its database.

A RADIUS client runs on the *Affirmed Networks Mobile Content Cloud* (MCC) device and sends RADIUS messages to the centralized RADIUS server over RADIUS interfaces as defined in RFC 2865, RFC 2866, and RFC 3162 for RADIUS Authentication, Authorization and Accounting.

RADIUS Interface Support for CUPS

The MCC Control and User Plane Separation (CUPS) architecture supports the RADIUS interface. The RADIUS interface is configured between the PGW-C and the AAA server. There is no change to the RADIUS interface configuration.

The PGW-C sends authentication, authorization, and accounting information over PFCP to the PGW-U via the Sx interface using vendor-specific information elements (IEs). The PGW-U sends traffic usage information to the PGW-C via the Sx interface. The PGW-C then reports the usage information to the AAA server via the RADIUS interface. See the *Affirmed Networks Gateway Administration Guide* for more information on the MCC CUPS architecture.

About RADIUS-Client Server Groups and Peers

Each RADIUS-client server group contains a list of peers (remote RADIUS servers) with which the system communicates. Each peer entry contains a server-group name, peer name, local loopback IP address, local port, remote IP address, remote port, secret key, and associated local network-context.

You can use the same local loopback IP address for multiple peers across multiple RADIUS-client server groups, but you configure each peer with a unique local UDP port to serve as the base port from which the system allocates a port range. All ports within the allocated range are reserved for that peer. Configuring multiple peers allows each peer to act as a backup for another. If one peer within the server group is unavailable, the RADIUS client (running on the MCC device) tries to connect with another peer within the group. You must configure separate server groups for RADIUS authorization services and RADIUS accounting services.

Note: When using the same IP address for multiple ports within a RADIUS-client server group, ensure that the base ports are separated by a minimum range of 512. The port range is calculated using the following formula: (number of maximum allowed slots * 4).

VRF-Lightweight and Normal Network-Contexts

As part of network context scaling, a RADIUS peer can be configured with either Virtual Routing and Forwarding (VRF) lightweight network contexts or normal network contexts.

RADIUS peers associated with VRF lightweight network contexts and RADIUS peers associated with normal network contexts are configurable within the same RADIUS-client server group or across RADIUS-client server groups.

Network context scaling for RADIUS peers configured with VRF lightweight network contexts is CPU intensive and involves a significant number of message transactions that may impact GwServiceDirector, NetDirector, NetAgent, NAS, PDNGW, SGW, RADIUS, and FastPath tasks. Affirmed Networks recommends scaling the configuration in a controlled manner or diverting traffic only after all of the new configurations have successfully completed.

	VRF-Lightweight Network Context	Normal Network Context
Maximum Number of Network Contexts	2000	128
Maximum Number of RADIUS-Client Server Groups	4000	770
Maximum Number of RADIUS Peers per Cluster	12,000	770*
Maximum Number of RADIUS peers per RADIUS-Client Server Group	256	256

* See [Configuring the Number of RADIUS Ports Supported Per RADIUS Peer \(page 1-5\)](#) for more information.

Note: The VRF-lightweight network context type does not apply to RADIUS-server server groups.

Configuring the Number of RADIUS Ports Supported Per RADIUS Peer

By default, each RADIUS peer supports four RADIUS client ports in addition to one port for net health. You can configure the number of RADIUS client ports supported per RADIUS peer using the following command.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group radius-client server-group <server-group-name>
peer <peer-name> num-local-ports <value>
```

num-local-ports – Specifies the number of RADIUS client ports per RADIUS peer. Possible values are 1 to 4. The default value is 4. The value configured for the **num-local-ports** parameter is in addition to one port used for net health.

For RADIUS peers configured with normal network contexts, when adding more RADIUS peers, the **num-local-ports** value has to be reduced to meet the below criteria.

$$\text{Total peers} * (\text{num-local-ports} + 1) < 3000$$

For example, if there are 500 RADIUS peers in a cluster and **num-local-ports** = 4, you would not need to lower the **num-local-ports** value because the number of RADIUS peers * the number of RADIUS client ports is less than 3000:

$$500 * (4 + 1) = 2500$$

However, if there are 700 RADIUS peers in a cluster and **num-local-ports** = 4, you would need to lower the **num-local-ports** value because the number of RADIUS peers * the number of RADIUS client ports exceeds 3000:

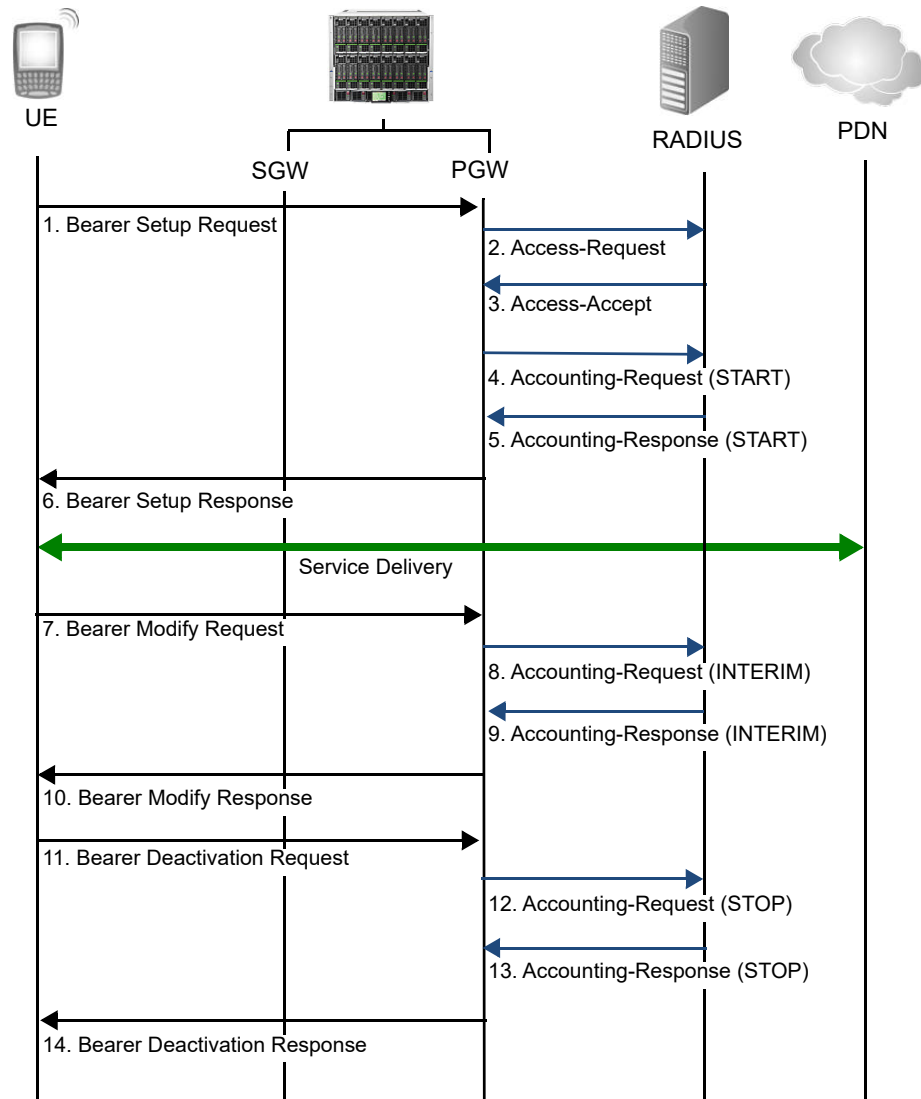
$$700 * (4 + 1) = 3500$$

Each RADIUS client port supports 256 outstanding transactions, so when you lower the number of RADIUS client ports supported per peer (**num-local-ports**), you reduce the number of outstanding transactions supported per peer.

RADIUS Message Flow

Figure 1-1 illustrates the basic message flow for contacting the RADIUS Server.

Figure 1-1 Basic Message Flow for Contacting the RADIUS Server



- 1 The user equipment (UE) sends a Bearer Setup Request message to the MCC GGSN/PGW containing the username and password.
- 2 The GGSN/PGW sends an Access-Request message to the RADIUS server containing the username, encrypted password and shared key.
- 3 The RADIUS server authenticates and authorizes the user. If the authentication succeeds, the RADIUS server returns an Access-Accept message containing the subscriber's authorization information, such as the allocated IPv4/IPv6 address.

Note: If the RADIUS server cannot match the user credentials against those stored in its database, it returns an Access-Reject message and the GGSN/PGW terminates the session.

- 4 The GGSN/PGW sends the RADIUS server an Accounting-Request START message to signal the start of the user session. This message contains parameters, such as the user ID, IPv4 address and/or the IPv6 prefix used by application servers to identify the user.
- 5 The RADIUS server responds with an Accounting-Response message to acknowledge the start of the user session.
- 6 The GGSN/PGW responds to the UE with a Bearer Setup Response message and service delivery to the UE begins.
- 7 If parameters change during an active session, such as a RAT change, SGSN/SGW address change, or QoS change, the UE sends a Bearer Modify Request message to the GGSN/PGW.
- 8 Upon receipt of a Bearer Modify Request, the GGSN/PGW sends an Accounting-Request INTERIM message to the RADIUS server.
- 9 The RADIUS server responds with an Accounting-Response to the GGSN/PGW to acknowledge the successful receipt of the Accounting-Request INTERIM message.
- 10 The GGSN/PGW responds to the UE with a Bearer Modify Response message.
- 11 To close the session, the subscriber sends a Bearer Deactivation Request message to the GGSN/PGW.
- 12 The GGSN/PGW sends the RADIUS server an Accounting-Request STOP message.
- 13 The RADIUS server deallocates the IPV4 address and/or IPv6 prefix initially allocated to the subscriber and sends the GGSN/PGW an Accounting-Response STOP message.
- 14 The GGSN/PGW responds to the UE with a Bearer Deactivation Response message to terminate the session.

RADIUS Messaging Following Geo-Redundancy Switchover

When a geo-redundancy switchover occurs, all sessions carrying RADIUS Accounting messages expire interim update timers simultaneously. This results in a wave of signaling messages for all active sessions because the RADIUS timers all start simultaneously (on the switchover). With the distribute RADIUS messaging burst feature, the distribution of RADIUS Accounting messages is normalized after a geo-redundancy switchover and prevents a wave of RADIUS accounting records being sent to the RADIUS server. For more information on Geographic Redundancy, see the *Acuitas User's Guide*.

RADIUS Configuration

This section describes the steps for configuring an MCC gateway to communicate with a RADIUS server via the RADIUS authorization and accounting interfaces.

To configure an MCC gateway to use RADIUS for authentication, authorization, and accounting, perform the following tasks:

- 1 Configure RADIUS-client server groups and remote RADIUS server peers ([page 1-9](#)).
- 2 Configure the RADIUS authorization service ([page 1-15](#)).
- 3 Configure the RADIUS accounting service ([page 1-18](#)).
- 4 Associate the RADIUS authorization and accounting services with a Control Profile ([page 1-24](#)).
- 5 Associate the Control Profile with a Subscriber Analyzer ([page 1-24](#)).

Configuring RADIUS-Client Server Groups and Remote Peers

A RADIUS-client server group contains a list of remote RADIUS server peers with which the MCC gateway communicates. You must configure separate server groups for RADIUS authorization services and RADIUS accounting services.

- 1 Configure a server group for RADIUS authorization services and a separate server group for RADIUS accounting services.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group radius-client server-group
<radius-auth-server-group-name>
```

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group radius-client server-group
<radius-acct-server-group-name>
```

The following table describes the configurable parameters.

Parameter	Description
dscp-value	Specifies the Differentiated Services Code Point (DSCP) value with which to mark all IP packets on this RADIUS server group. For more information, see Configuring DSCP Values for Control Plane Traffic (page 1-43) .
geo-redundant-group	Specifies the Geographic Redundancy Group. The Geographic Redundancy Group must already exist. For information on configuring a Geographic Redundancy Group, see the Configuring Geo-Redundancy Health Checks on RADIUS Interfaces (page 1-44) .
nas-identifier	Specifies the hostname of the GGSN/PGW requesting user authentication used in RADIUS Access-Request and Accounting-Request messages. <i>Note:</i> If you do not configure the <i>NAS-Identifier</i> , the MCC gateway sends a string form of the remote RADIUS server peer's local loopback address in the <i>NAS-Identifier</i> attribute. <i>Note:</i> When using the <i>nas-identifier</i> parameter under the <i>APN</i> object, do not configure the <i>nas-identifier</i> parameter under the RADIUS server group.
nas-ip-address	Specifies the IP address of the GGSN/PGW requesting user authentication used in RADIUS Access-Request and Accounting-Request messages. <i>Note:</i> If you do not configure the <i>NAS-IP-Address</i> , the MCC gateway sends the remote RADIUS server peer's local loopback address in the <i>NAS-IP Address</i> attribute.

Parameter	Description
peer-retry-mode	<i>Note: Disable the administrative state of all RADIUS server peers in the RADIUS server group before modifying this parameter.</i> Select one of the following modes: ■ Sequential – If the MCC gateway does not receive a response from a remote RADIUS server peer within the configured time limit, it retransmits the request to the <i>same</i> peer (until the request retry limit is reached) before selecting another active peer in the same group (in a round-robin fashion). ■ Alternate – (default) If the MCC gateway does not receive a response from a remote RADIUS server peer within the configured time limit, it retransmits the request to the <i>next</i> active peer in the same group (in a round-robin fashion). <i>When this mode is selected, the configured priority does not apply.</i> ■ alternate-with-priority – All new requests are sent to the RADIUS server peer with the highest priority (1 being the highest). If the RADIUS server peer with the highest priority is unavailable, the request is sent to the peer with the next highest priority in the same group. If multiple RADIUS server peers are assigned the highest priority (1), the RADIUS client (running on the MCC gateway) uses a round-robin approach to select from among those peers for new requests. <i>Note: See Failover Message Flows (page 1-68) for more information on these modes.</i>
net-health-report-enabled	Specifies whether to monitor network health towards the interface: ■ true ■ false <i>Note: If RADIUS is used for Geographic Redundancy health check, set this parameter to <code>true</code>. For Geographic Redundancy configuration information, see Configuring Geo-Redundancy Health Checks on RADIUS Interfaces (page 1-44).</i>
link-type	Specifies whether the link used to monitor neighboring network elements is a private (local) or shared IP address among the primary and secondary network elements: ■ private ■ shared
remote-health-check	Enable or disable Enhanced RADIUS Health Check (ERHC) used to monitor the status of remote RADIUS server peers within a RADIUS server group. ■ true – ERHC is enabled. ■ false – (default) ERHC is disabled. <i>Note: For more information on ERHC, see Enhanced RADIUS Health Check (page 1-66).</i>

Parameter	Description
<code>remote-health-check-period</code>	Specifies the number of seconds between remote RADIUS Authentication/Accounting Health Check messages. The default is 15. The range is 1-300 seconds.
<code>remote-health-check-response-period</code>	Specifies the number of seconds the RADIUS client (running on the MCC gateway) waits for a response after sending a RADIUS Authentication/Accounting Health Check message to the remote RADIUS server peer. The default is 1. The range is 1-10 seconds.
<code>remote-health-check-retry-count</code>	Specifies the maximum number of health check attempts to a RADIUS server peer before declaring the peer down. The default is 3. The range is 0-10.
<code>response-expected</code>	<i>Note: This parameter applies when mirroring RADIUS accounting messages to a second RADIUS accounting server. For more information on RADIUS mirroring, see Mirroring RADIUS Accounting Messages (page 1-37).</i> Specifies whether the MCC gateway should expect a response from the mirrored RADIUS accounting server(s) (server to which mirrored messages are sent) in this RADIUS server group. ■ true – (default) The MCC gateway expects to receive a response from the RADIUS accounting servers in this RADIUS server group. ■ false – The MCC gateway does not expect to receive a response from the RADIUS accounting servers in this RADIUS server group and as a result, the RADIUS accounting servers remain in an operationally “Up” state. <i>Note: When this parameter is set to false, RADIUS Health Check (<code>remote-health-check</code>) must be disabled for this RADIUS server group.</i>
<code>server-group-type</code>	<i>Note: Disable the administrative state of all RADIUS server peers in the RADIUS server group before modifying this parameter.</i> Specifies the type of RADIUS server group. ■ NotApplicable ■ Authorization ■ Accounting
<code>session-aware-peer-selection</code>	Specifies whether to support session aware peer selection: ■ true – All RADIUS Accounting-Requests for a given session are sent to the same RADIUS server peer. If the original peer is unavailable, the system transmits subsequent RADIUS accounting messages to another active peer in the same RADIUS server group, selected in a round-robin fashion. The PGW continues to send the remaining RADIUS transactions for that call on the newly selected peer even if the original peer comes back up. ■ false – RADIUS Accounting-Requests for a given session are sent to peers selected in a round-robin fashion.

- 2 Use the following command to add peers to the RADIUS authentication and accounting server groups.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group radius-client server-group
<radius-server-group-name> peer <radius-peer-name>
```

The following table describes the configurable parameters.

Parameter	Description
admin-state	Specifies the administrative state of the RADIUS server peer connection. ■ enabled ■ disabled
disconnect-coa-local-server-port	Note: Disable the administrative state of the remote RADIUS server peer before modifying this parameter. Specifies the local (RADIUS client) UDP port number to which Disconnect-Request messages are sent. The default is 3799. Note: Ensure that the disconnect local server port does not conflict (same IP and same port) with the local port range of any of the configured RADIUS server peers. Note: If multiple RADIUS server peers have the same local IP and the same disconnect local server port, only one RADIUS disconnect server is instantiated.
disconnect-coa-server-admin-state	Note: Disable the administrative state of the remote RADIUS server peer before modifying this parameter. Specifies whether the PGW/GGSN receives and processes RADIUS Disconnect-Request messages from the remote RADIUS server. ■ enabled ■ disabled (default) Note: To use the Change-of-Authorization (CoA) functionality, you must configure each RADIUS server peer within a RADIUS server group with a unique peer name.
local-port	Specifies the local (RADIUS client) UDP port number. Note: When using the same IP address for multiple ports within a RADIUS-client server group, ensure that the base ports are separated by a minimum range of 512. The port range is calculated using the following formula: (number of maximum allowed slots * 4).
loopback-ip	Specifies the signaling port IP address.

Parameter	Description
<code>max-num-response-timeout</code>	Specifies the maximum number of retransmission attempts allowed for a server group after which the RADIUS client (running on the MCC gateway) rejects the request. Enter a value between 1-5 (default is 3). Note: <i>Affirmed Networks recommends that all RADIUS server peers within a server group use the same value for this parameter. If the peers use different values, any of the values may apply.</i>
<code>num-local-ports</code>	Specifies the number of RADIUS client ports per RADIUS peer. Possible values are 1 to 4. The default value is 4. The value configured for the <code>num-local-ports</code> parameter is in addition to one port used for net health. When adding more RADIUS peers, the <code>num-local-ports</code> value has to be reduced to meet the below criteria. Total peers * (num-local-ports + 1) < 3000 Note: See About RADIUS-Client Server Groups and Peers (page 1-3) for more information.
<code>network-context</code>	Specifies the network context containing the RADIUS interface. The network context must already exist. For information on configuring a network context, see the <i>Affirmed Networks Gateway Administration Guide</i>.
<code>pacer-max-transactions-per-interval</code>	Specifies the maximum number of transactions sent to a RADIUS peer per time interval. This parameter is only visible in the CLI when set to a non-zero value. Use the no form of this command to disable this parameter. The default is 256. Note: <i>The value configured for the <code>pacer-max-transactions-per-interval</code> will be shared between multiple CSMs in one cluster.</i>
<code>pacer-time-interval</code>	Specifies the RADIUS transaction time interval, in milliseconds, after which the pacer resets its outgoing transaction count. The default is 100 milliseconds. Note: <i>This parameter is only visible in the CLI when the <code>pacer-max-transaction-per-interval</code> parameter is set to a non-zero value.</i>
<code>priority</code>	Specifies the RADIUS server peer priority within the RADIUS server group (with 1 being the highest priority). Note: <i>Applies only when the <code>peer-retry-mode</code> parameter is set to <code>alternate-with-priority</code>.</i>
<code>remote-ip-address</code>	Specifies the remote (RADIUS server) IP address.

Parameter	Description
<code>remote-port</code>	Specifies the remote (RADIUS server) UDP port number. <i>Note: The default remote authorization port is 1812 and the default remote accounting port is 1813.</i>
<code>secret-key</code>	Specifies the secret key string shared for a connection. <i>Note: The exclamation point (!) is a special character used to comment out the rest of the CLI command. If a CLI command requires the use of !, Operators should add \ before ! to mask this behavior. For example, \!password.</i>
<code>server-dead-num-attempts</code>	Specifies the number of consecutive unanswered requests allowed for a peer before the system declares a RADIUS server peer down. <i>Note: Setting this parameter to a value lower than 500 may result in the RADIUS server peer frequently being taken out of service at very high call rates.</i>
<code>server-dead-timeout-value</code>	Specifies the number of seconds to wait before attempting to reconnect to the non-responsive RADIUS server peer after moving the peer from a pending state to an inactive state.
<code>server-resp-timeout-value</code>	Specifies the number of seconds to wait before the system retransmits the request. Enter a value between 1-5 seconds (default is 1 second).
<code>server-type</code>	Specifies the type of RADIUS server peer. ■ Primary – (default) Active peers used for RADIUS Authorization and Accounting services. ■ Secondary – Used for RADIUS Authorization and Accounting services, only when the primary peers are down. <i>Note: This parameter is available only when the <code>peer-retry-mode</code> parameter is set to <i>Alternate</i> or <i>Sequential</i>.</i>

Configuring the RADIUS Authorization Service

Perform the following tasks to configure the RADIUS Authorization service on an MCC gateway.

- 1 Configure the data record templates to filter attributes in RADIUS Access-Request and Access-Accept messages.

The Access-Request and Access-Accept data record templates are used to customize messages transmitted to/from the RADIUS authorization interface. Data record templates allow you to include or exclude certain optional/conditional attributes in the messages.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|ggsn] auth radius [access-accept|access-request] <template-name>
<attribute-name> presentFlag [conditional|no|yes]
```

where

Parameter	Description
[access-accept access-request]	Specifies the type of data record template.
<template-name>	Specifies the name of the data record template.
<attribute-name>	Specifies the name of the AVP you are modifying.
presentFlag	Access-Accept templates – Options include: ■ no – The MCC gateway discards this attribute when received in the Access-Accept message. ■ yes – The MCC gateway processes this attribute when received in the Access-Accept message. Access-Request templates – Options include: ■ conditional – The MCC gateway sends the attribute in the Access-Request message if it is available. ■ no – The MCC gateway does not include the attribute in the Access-Request message. ■ yes – The MCC gateway includes the attribute in the Access-Request message.

Use the following CLI command to list the configurable attributes in the Access-Accept and Access-Request data record templates:

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|ggsn] auth radius [access-accept|access-request]
<template-name> ?
```

- 2 Configure the RADIUS authorization interface and associate the RADIUS-client server group and Access-Request/Access-Accept data record templates with this service.

```
an3000-mcm-slot7cpu1(config)# service-construct interface  
authorization-service <auth-interface-name> admin-state enabled
```

where <auth-interface-name> is the name of the RADIUS authorization interface.

The following table describes the configurable parameters.

Parameter	Description
acct-session-id-format	Specifies whether to include/exclude the PDN session start time in the RADIUS Acct-Session-Id and Acct-Multi-Session-Id AVPs sent in RADIUS Access-Request messages. ■ default – (default) The PGW/GGSN does not include the PDN session start time in the Acct-Session-Id and Acct-Multi-Session-Id AVPs sent in RADIUS Access-Request messages. (Acct-Session-ID Format = primary GW IP address + bearer-ID + 0xFF + subscriber session ID) ■ include-timestamp – The PGW/GGSN includes the PDN session start time in the Acct-Session-Id and Acct-Multi-Session-Id AVPs sent in RADIUS Access-Request messages. (Acct-Session-ID Format = primary GW IP address + bearer ID + 0xFF + timestamp + acct session ID Seq num(if != 0) + 0xFF + subscriber session ID) <i>Note: The Acct-Session-Id and Acct-Multi-Session-Id must be consistent across authorization and accounting messages. Therefore, when the Workflow Control Profile has both a default authorization and default accounting interface configured, the acct-session-id-format configured for the accounting-service takes precedence and applies to the Acct-Session-Id and Acct-Multi-Session-Id AVPs in both the Accounting and Authorization messages.</i>
apn-reporting	<i>Applies to the PGW only.</i> Specifies the Access Point Names (APNs) to be reported over the RADIUS accounting interface. ■ original – Initial APN ■ mapped – Final / current APN ■ prior-mapped – APN mapped prior to the mapped (final/current) APN. This setting allows APN Reporting to continue to be based on the previously-mapped APN (even following a remap to a final APN) The apn-reporting value configured at the interface level overrides the apn-reporting value configured for the Control-Profile and is used to fill the Called-Station-ID AVP and User-Name AVP sent in messages to external devices. For more information, see Configuring APN Reporting at the Interface Level (page 1-36). <i>Note: The interface-level apn-reporting parameter does not support the custom option.</i>

Parameter	Description
framed-route-default-metric	Specifies a default metric for Framed-Routes with missing metrics or with a metric of 0. If this parameter is configured, the PGW accepts a Framed-Route AVP with a missing metric or a metric of 0 and applies the configured metric to it. If this parameter is not configured, the PGW ignores a Framed-Route AVP with a missing metric or a metric of 0 and the <code>show pdn session</code> CLI output will not contain the corresponding Framed-Route entry. By default, this parameter is not configured.
ignore-radius-failure	Controls whether the GGSN/PGW processes a RADIUS failure response or ignores the failure and continues with the Create PDP Context/Session procedure. See Configuring the ignore-radius-failure Parameter (page 1-53) for more information. ■ true – The GGSN/PGW ignores the RADIUS (authorization/accounting) failure response and continues with the Create PDP Context/Session procedure. ■ false – (default) The GGSN/PGW processes the RADIUS (authorization/accounting) failure response and rejects the Create PDP Context/Session Request. <i>Note: If the failure is a RADIUS authorization failure, MCC uses the local configuration (if configured) to create the PDP Context/Session.</i> <i>Note: Changes to this parameter take effect immediately.</i>
server-group	Specifies the RADIUS authorization server group name. <i>Note: The RADIUS authorization server group must already exist. See Configuring RADIUS-Client Server Groups and Remote Peers (page 1-9).</i>
nas-port-type	Specifies the NAS-Port-Type attribute sent in RADIUS Authorization messages. ■ virtual ■ wireless-other (default)
[pgw ggsn] Associate the Access-Request/Access-Accept data record templates with this service:	
[access-accept-record access-req-record]	Specifies the type of data record template.
<template-name>	Specifies the name of the data record template. <i>Note: The data record templates must already exist.</i>

Configuring the RADIUS Accounting Service

Perform the following tasks to configure the RADIUS accounting service on an MCC gateway.

- 1 Configure the data record templates applied to RADIUS Accounting-Request (Start/Stop/Interim) messages.

The Accounting-Request (Start/Stop/Interim) data record templates are used to customize messages transmitted to/from the RADIUS accounting interface. Data record templates allow you to include or exclude certain optional/conditional attributes in the messages.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pdngw accounting radius [start|interim|stop] <template-name>
<attribute-name> presentFlag [conditional|no|yes]
```

where

Parameter	Description
[start interim stop]	Specifies the type of data record template.
<template-name>	Specifies the name of the data record template.
<attribute-name>	Specifies the name of the AVP you are modifying.
presentFlag	Options include: ■ conditional – The MCC gateway sends the attribute in the Accounting-Request message if it is available. ■ no – The MCC gateway does not include the attribute in the Accounting-Request message. ■ yes – The MCC gateway includes the attribute in the Accounting-Request message.

Use the following CLI command to list the configurable attributes in the RADIUS Accounting-Request (Start/Stop/Interim) data record templates:

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pdngw accounting radius [start|interim|stop] <template-name> ?
```


- 2 Configure the RADIUS accounting event trigger template to specify the events that trigger an interim update:

```
an3000-mcm-slot7cpu1(config)# service-construct
interface-trigger-template event-trigger-template <template-name>
```

The following table describes the configurable parameters.

Parameter	Description
qos-change-close-trigger	Specifies whether a QoS Change triggers an update. Value 0 disables the trigger, value 1 - 20 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of QoS Change events to trigger a CDR close.
rat-type-change-close-trigger	Specifies whether a RAT Type Change triggers an update. Value 0 disables the trigger, value 1 - 20 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of RAT Type Change events to trigger a CDR close.
serving-node-change-close-trigger	Specifies whether a Serving Node Change triggers an update. Value 0 disables the trigger, value 1 - 20 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of Serving Node Change events to trigger a CDR close. <i>Note: Applies to the GGSN/PGW when the context is transferred to a new SGW/SGSN or to the SGW when the Mobility Management Entity (MME) is changed.</i>
serving-node-plmn-change-close-trigger	Specifies whether a Serving Node PLMN Change triggers an update. Value 0 disables the trigger, value 1 - 20 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of Serving Node PLMN Change events to trigger a CDR close. <i>Note: Applies to PGW and GGSN only.</i>
time-limit-close-trigger	Applies to offline interfaces only. If interim updates are enabled on the interface, specifies the number of events to trigger a CDR close. Enter a value between 1-20.
ue-time-zone-change-close-trigger	Specifies whether a UE Time Zone Change triggers an update. Value 0 disables the trigger, value 1 - 20 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of UE Time Zone Change events to trigger a CDR close.
user-location-change-close-trigger	Specifies whether a User Location Change triggers an update. Value 0 disables the trigger, value 1 - 20 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of User Location Change events to trigger a CDR close.

Parameter	Description
volume-limit-close-trigger	Applies to offline interfaces only. If Data Volume Limit is enabled on the interface, specifies the number of events to trigger a CDR close. Enter a value between 1-20.
user-csg-change-close-trigger	Specifies whether a CSG change triggers an update. Value 0 disables the trigger, value 1 - 20 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CSG change events to trigger a CDR close.
change-in-up-to-ue-close-trigger	Applies to offline interfaces only. Specifies whether a change in the User Plane (UP) to User Equipment (UE) (for example, a switch between S11-U and S1-U) triggers an update. Value 0 disables the trigger, value 1 - 20 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of UP to UE Change events to trigger a CDR close. <i>Note: Applies to the SGW only.</i>

3 Configure the RADIUS accounting interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
accounting-service <acct-interface-name> admin-state enabled
```

where **<acct-interface-name>** is the name of the RADIUS accounting interface.

The following table describes the configurable parameters.

Parameter	Description
acct-session-id-format	Specifies whether to include/exclude the PDN session start time in the RADIUS Acct-Session-Id and Acct-Multi-Session-Id AVPs sent in RADIUS Accounting-Request messages. ■ default – (default) The PGW/GGSN does not include the PDN session start time in the Acct-Session-Id and Acct-Multi-Session-Id AVPs sent in RADIUS Access-Request messages. (Acct-Session-ID Format = primary GW IP address + bearer-ID + 0xFF + subscriber session ID) ■ include-timestamp – The PGW/GGSN includes the PDN session start time in the Acct-Session-Id and Acct-Multi-Session-Id AVPs sent in RADIUS Access-Request messages. (Acct-Session-ID Format = primary GW IP address + bearer ID + 0xFF + timestamp + acct session ID Seq num(if != 0) + 0xFF + subscriber session ID) <i>Note: The Acct-Session-Id and Acct-Multi-Session-Id must be consistent across authorization and accounting messages. Therefore, when the Workflow Control Profile has both a default authorization and default accounting interface configured, the acct-session-id-format configured for the accounting-service takes precedence and applies to the Acct-Session-Id and Acct-Multi-Session-Id AVPs in both the Accounting and Authorization messages.</i>
apn-reporting	Applies to the PGW only. Specifies the Access Point Names (APNs) to be reported over the RADIUS accounting interface. ■ original – Initial APN ■ mapped – Final / current APN ■ prior-mapped – APN mapped prior to the mapped (final/current) APN. This setting allows APN Reporting to continue to be based on the previously-mapped APN (even following a remap to a final APN) The apn-reporting value configured at the interface level overrides the apn-reporting value configured for the Control-Profile and is used to fill the Called-Station-ID AVP and User-Name AVP sent in messages to external devices. For more information, see Configuring APN Reporting at the Interface Level (page 1-36). <i>Note: The interface-level apn-reporting parameter does not support the custom option.</i>

Parameter	Description
<code>delay-failure-handling-until-all-responses</code>	Specifies whether the GGSN/PGW waits for the Accounting-Response from all of the RADIUS servers to which an Accounting-Request START message was sent before processing a RADIUS failure response. ■ true – The GGSN/PGW waits for the Accounting-Response from all of the RADIUS servers to which an Accounting-Request START message was sent before processing a RADIUS accounting failure. After receiving all Accounting-Request START responses, the GGSN/PGW sends an Accounting-Request STOP message to all RADIUS servers, except the RADIUS servers for which the Accounting-Request START message failed. ■ false – The GGSN/PGW processes the RADIUS accounting failure response immediately and sends Accounting-Request STOP messages to only those servers from which it had received an Accounting-Response START message and clears the session. The <code>delay-failure-handling-until-all-responses</code> parameter can only be configured for the primary RADIUS accounting interface and will be applicable for all mirrored RADIUS accounting interfaces configured in the Control Profile, with which the primary RADIUS accounting interface is associated. See Processing RADIUS Failure Responses (page 1-53) for more information. <i>Note: Changes to this parameter take effect immediately.</i>
<code>is-ofc-enabled</code>	Controls whether the system stores accounting messages locally to the disk when the remote RADIUS server is unavailable. When the remote server becomes available, the system sends the pending accounting messages. ■ true ■ false (default)
<code>ignore-radius-failure</code>	Controls whether the GGSN/PGW processes a RADIUS failure response or ignores the failure and continues with the Create PDP Context/Session procedure. See Configuring the ignore-radius-failure Parameter (page 1-53) for more information. ■ true – The GGSN/PGW ignores the RADIUS (authorization/accounting) failure response and continues with the Create PDP Context/Session procedure. ■ false – (default) The GGSN/PGW processes the RADIUS (authorization/accounting) failure response and rejects the Create PDP Context/Session Request. <i>Note: If the failure is a RADIUS authorization failure, MCC uses the local configuration (if configured) to create the PDP Context/Session.</i> <i>Note: Changes to this parameter take effect immediately.</i>

Parameter	Description
wait-for-accounting-response	Controls whether the GGSN/PGW waits for an Accounting-Response from the RADIUS accounting server before sending the Create/Delete PDP Context Response or Create/Delete Session Response. See Processing RADIUS Accounting-Responses (page 1-52) for more information. ■ true – (default) The GGSN/PGW waits for an Accounting-Response before sending a Create/Delete PDP Context Response or Create/Delete Session Response to the SGSN/SGW. ■ false – The GGSN/PGW sends the Create/Delete PDP Context Response or Create/Delete Session Response to the SGSN/SGW without waiting for an Accounting-Response. <i>Note: Changes to this parameter take effect immediately.</i>
nas-port-type	Specifies the NAS-Port-Type attribute sent in RADIUS Accounting messages. ■ virtual ■ wireless-other (default)
pdngw	Configure interim updates on the primary accounting interface:
interim-update-interval	Specifies the number of seconds that the gateway waits before generating a new accounting record.
interim-updates	Specifies whether to enable or disable interim updates to the RADIUS server. ■ enabled ■ disabled
data-volume-limit	Specifies the number of kilobytes that the gateway maintains before generating a new accounting record. The volume usage counts are reset when an accounting record is closed and sent. The default is 0 (disabled).
event-trigger-template	Specifies the name of the event trigger template associated with the RADIUS accounting interface.

- 4 Associate the RADIUS-client server group and Accounting-Request data record templates with this service.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
accounting-service <acct-service-interface-name> pdngw radius
server-group <radius-server-group-name>
[interim-template|start-template|stop-template] <template-name>
```

where

Parameter	Description
<code>accounting-service</code>	Specifies the RADIUS accounting interface name.
<code>radius server-group</code>	Specifies the RADIUS accounting server group name. <i>Note: The RADIUS accounting server group must already exist.</i>
<code>[interim-template start-template stop-template]</code>	Specifies the type of data record template.
<code><template-name></code>	Specifies the name of the data record template. <i>Note: The data record templates must already exist.</i>

Completing the RADIUS Configuration Process

Perform the following steps to complete the RADIUS configuration process.

- 1 Associate the RADIUS authorization and accounting services with a Workflow Control Profile. The Workflow Control Profile must already exist.
- 2 Associate the Control Profile with a Subscriber Analyzer. The Subscriber Analyzer must already exist.

The Workflow Subscriber Analyzer enables Operators to identify and filter subscribers based on a set of attributes defined in the associated Control and Data Profiles. The Control Profile specifies the back-end interface to use for RADIUS. For information on performing these tasks, see the *Affirmed Networks Gateway Administration Guide*.

Optional RADIUS Configuration Tasks

This section describes optional RADIUS configuration tasks.

[Configuring RADIUS IP Pool Allocation \(page 1-26\)](#)

[Configuring RADIUS Accounting-Request ON/OFF Messages \(page 1-28\)](#)

- [Configuring a Time Delay Between Accounting-Request OFF/ON Messages \(page 1-29\)](#)
 - [Including/Excluding the Called-Station-ID in Accounting-Request ON/OFF Messages \(page 1-29\)](#)
-

[Configuring a RADIUS Server on a per-APN Basis Using APN Mapping \(page 1-30\)](#)

[APN Mapping/Redirection Using AAA Authentication \(page 1-31\)](#)

[Setting Up L2TP LAC functionality via RADIUS APN Mapping \(page 1-32\)](#)

[Establishing L2TP Tunnel via RADIUS Access-Accept \(page 1-33\)](#)

[Customizing the APN Reported on External Interfaces \(page 1-35\)](#)

[Configuring APN Reporting at the Interface Level \(page 1-36\)](#)

[Mirroring RADIUS Accounting Messages \(page 1-37\)](#)

[Configuring DSCP Values for Control Plane Traffic \(page 1-43\)](#)

[Configuring Geo-Redundancy Health Checks on RADIUS Interfaces \(page 1-44\)](#)

[Reporting Rating-Group Level Usage Statistics \(page 1-49\)](#)

[Suppressing NULL Characters in PCO PAP Username/Password Fields \(page 1-50\)](#)

[Configuring a NAS-Identifier per APN \(page 1-51\)](#)

[Processing RADIUS Accounting-Responses \(page 1-52\)](#)

[Processing RADIUS Failure Responses \(page 1-53\)](#)

[Processing the TMO-Virtual-Gi-ID VSA \(page 1-56\)](#)

Configuring RADIUS IP Pool Allocation

If the remote RADIUS server does not allocate an IP address to a user requesting access, you can configure the GGSN/PGW to allocate a UE IP address locally from either the remote UE Pool or local UE Pool.

Note: A quarantine timer is not applied to UE IP addresses allocated from the remote UE Pool.

Use the following command to configure RADIUS IP Pool Allocation:

```
an3000-mcm-slot7cpu1(config)# workflow control-profile  
<control-profile-name> ue-ip-allocation-mode radius-ue-ip-allocation  
remote-ip-alloc-fallback-option  
[alloc-none|alloc-local-pool|alloc-remote-pool]
```

where

Parameter	Description
control-profile	Specify the Workflow Control Profile to which the RADIUS Authorization and Accounting interfaces are associated.
ue-ip-allocation-mode	Specify <code>radius-ue-ip-allocation</code> as the IP allocation mode for the sessions associated with this Control Profile.

Parameter	Description
<code>remote-ip-alloc-fallback-option</code>	Select one of the following values: ■ alloc-none – (default) The GGSN/PGW rejects the request if the remote RADIUS server fails to send the Framed-IP-Address attribute or IPv4/IPv6 pool names in the Access-Accept message. ■ alloc-local-pool – The GGSN/PGW allocates the UE IP address from the local pool if the remote RADIUS server fails to send the Framed-IP-Address attribute or IPv4/IPv6 pool names in the Access-Accept message and if configured, quarantines these locally-assigned IP addresses. For information on configuring a quarantine timer for the local UE pool, see the <i>Affirmed Networks Gateway Administration Guide</i>. ■ alloc-remote-pool – The GGSN/PGW allocates the UE IP address from the remote pool if the remote RADIUS server fails to send the Framed-IP-Address attribute or IPv4/IPv6 pool names in the Access-Accept message. <i>Note:</i> When this parameter is set to <code>alloc-remote-pool</code> and the RADIUS server fails to send the Framed IP Address or IPv4/IPv6 pool names, GGSN/PGW allocates an IP address from the remote UE Pool. Later, if the remote RADIUS server requests to reserve the same IP address, the GGSN/PGW rejects the request.

Configuring RADIUS Accounting-Request ON/OFF Messages

MCC supports RADIUS Accounting-Request ON/OFF messages to ensure session information between the GGSN/PGW and the remote RADIUS server is synchronized.

The GGSN/PGW sends Accounting-Request ON messages to the remote RADIUS server to indicate that the MCC device is up after a scheduled reboot. The GGSN/PGW sends Accounting-Request OFF messages to the remote RADIUS server to indicate that the MCC device is going down due to a scheduled reboot.

The Accounting-Request ON/OFF messages are sent on all the configured RADIUS server peer interfaces to that remote RADIUS server and for every GGSN/PGW and APN on which Accounting ON/OFF messages are enabled.

Use the following command to enable Accounting-Request ON/OFF messages:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>  
allow-accounting-on-off-msg [disabled|enabled]
```

where

Parameter	Description
zone	Specifies the zone name.
apn	Specifies the Access Point Name.
allow-accounting-on-off-msg	Specifies whether to allow the GGSN/PGW to send Accounting-Request ON/OFF messages to the remote RADIUS server. ■ enabled ■ disabled

Configuring a Time Delay Between Accounting-Request OFF/ON Messages

You can specify a time delay (in milliseconds) between Accounting-Request OFF/ON messages to ensure the RADIUS server synchronizes information with the MCC gateway before receiving the Accounting-Request ON. The range is 600000 - 1800000. By default, this parameter is not configured.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gateway-common <gateway-profile-name> radius-acct-off-on-delay <value>
```

When the `radius-acct-off-on-delay` timer is configured and the system reboots, the MCC checks if the Accounting-Request OFF message was sent prior to the reboot.

- If the Accounting-Request OFF message was sent prior to the reboot, the MCC sends the Accounting-Request ON message to the RADIUS server once the `radius-acct-off-on-delay` timer expires.
- If it was not sent prior to the reboot (for example, due to an ungraceful shutdown of the system), the MCC sends the Accounting-Request OFF message after the system reboots and starts the `radius-acct-off-on-delay` timer. Once the timer expires, the MCC sends the Accounting-Request ON message to the RADIUS server.

Use the `no` form of this command to disable this parameter.

Command: `no radius-acct-off-on-delay`

Including/Excluding the Called-Station-ID in Accounting-Request ON/OFF Messages

Use the following command to include/exclude the Called-Station-ID AVP in RADIUS Accounting-Request ON/OFF messages.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gateway-common <gateway-profile-name>
include-called-station-id-in-acct-on-off [true|false]
```

- **true** – (default) The MCC gateway includes the Called-Station-ID AVP in RADIUS Accounting-Request ON/OFF messages. The MCC gateway sends one Accounting-Request ON message per APN configured and per RADIUS peer.
- **false** – The MCC gateway excludes the Called-Station-ID AVP in RADIUS Accounting-Request ON/OFF messages. The MCC gateway sends only one Accounting-Request ON/OFF per gateway service regardless of the number of APNs configured for the GW service.

Configuring a RADIUS Server on a per-APN Basis Using APN Mapping

Operators can configure a RADIUS server peer on a per-APN (Gn-Gi APN) basis while using APN mapping. When this feature is enabled and the GGSN receives a Create PDP Context request message for a given APN, the GGSN sends the authentication and accounting information to the RADIUS server peer defined in the mapped APN Subscriber Analyzer and Workflow Control Profile parameters.

Set parameters for this feature using the following command:

```
an3000-mcm-slot7cpu1(config)# workflow control-profile <name>
apn-mapping-control explicit mapped-apn-name <name>
allow-mapped-apn-based-sub-analyser [false|true]
```

where

Parameter	Description
workflow control-profile	Specifies the Workflow Control Profile name for the generic APN. The generic APN must already exist. See the <i>Affirmed Networks Gateway Administration Guide</i> for information on configuring APNs. The Workflow Control Profile specifies the back-end interfaces to use for authentication and accounting.
apn-mapping-control	To map the generic APN to the APN specified in the mapped-apn-name parameter, specify explicit .
mapped-apn-name	Specifies the mapped APN. The mapped APN must already exist. See the <i>Affirmed Networks Gateway Administration Guide</i> for information on configuring APNs.
allow-mapped-apn-based-sub-analyser	Specifies one of the following options: ■ false – (default) Maps the generic APN to the specified mapped APN so that external entities, such as RADIUS and policy and charging servers, can provide differentiated treatments. ■ true – Maps the generic APN to the specified mapped APN and applies the Subscriber Analyzer rules associated with the mapped APN to the subscriber. In doing so, the MCC applies the Workflow Control Profile associated with the mapped APN Subscriber Analyzer to the subscriber.

APN Mapping/Redirection Using AAA Authentication

The MCC supports the ability to map to an APN using information received from the RADIUS server in the Called-Station-Id attribute and to use the mapped APN information to redirect the subscriber session to a new Control Profile.

When the PGW/GGSN receives an Access-Accept message containing a Called-Station-Id that is different from the one sent in the Access-Request, the PGW/GGSN maps the subscriber session to the APN received in the Called-Station-Id attribute. In doing so, the subscriber session uses the Subscriber Analyzer and Workflow Control Profile parameters defined for the APN received in the Called-Station-Id attribute.

To use RADIUS-based APN mapping/redirection, the APN received in the Called-Station-Id AVP must match an APN configured locally. If the APN received in the Called-Station-Id AVP, within the Access-Accept message, is not configured locally on the PGW, the subscriber session is rejected. If the Access-Accept message does not contain a new Called-Station-Id AVP, the subscriber session continues with the control profile selected initially.

To activate this feature, perform the following CLI commands.

- 1 Set the `apn-mapping-control` parameter to `external` to allow APN mapping from the RADIUS server.

```
an3000-mcm-slot7cpu1(config)# workflow control-profile <name>
apn-mapping-control external
```

Note: If `apn-mapping-control` parameter is not set to `external`, the PGW ignores the new Called-Station-Id AVP and continues the session with the control profile selected initially.

- 2 Set the `allow-mapped-apn-based-sub-analyzer` parameter to `true` to apply the Subscriber Analyzer rules associated with the APN received in the Called-Station-Id attribute to the subscriber.

```
an3000-mcm-slot7cpu1(config)# workflow control-profile <name>
allow-mapped-apn-based-sub-analyzer true
```

Note: Only one APN redirection is allowed per subscriber.

Setting Up L2TP LAC functionality via RADIUS APN Mapping

MCC supports the ability to map to an APN using information received from the RADIUS server in the Called-Station-Id and to use the mapped APN information to set up Layer 2 Tunneling Protocol (L2TP) functionality.

This feature requires the PGW/GGSN to send two RADIUS Authentication messages; one for APN mapping through the Called-Station-Id attribute and another to request L2TP tunneling attributes.

When the PGW/GGSN receives an Access-Accept message containing a Called-Station-Id that is different from the one sent in the first Access-Request, the PGW/GGSN maps the subscriber session to the specified mapped APN received in the Called-Station-Id attribute. In doing so, the subscriber session uses the Subscriber Analyzer and Workflow Control Profile parameters defined for the mapped APN.

Note: To ensure the subscriber session uses the mapped APN Subscriber Analyzer and Workflow Control Profile parameters, set `apn-mapping-control` to `external` and set `allow-mapped-apn-based-sub-analyzer` to `true` in the Workflow Control Profile.

If the `ue-ip-allocation-mode` in the Workflow Control Profile for the mapped APN is set to `lms-ue-ip-allocation`, the PGW/GGSN sends a second Access-Request message to the RADIUS server defined in the mapped APN Subscriber Analyzer and Workflow Control Profile parameters. Once the PGW/GGSN receives the Access-Accept message with L2TP Tunnel attributes, the existing L2TP functionality for establishing the L2TP tunnel and session with the L2TP Network Server (LNS) and PPP PAP/CHAP negotiation for UE IP address allocation is executed.

This feature introduces two configuration parameters under the APN object.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <name>
pco-username-type [realm-only|username-plus-realm]
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <name>
user-password-type [default-password|pco-password]
```

- **pco-username-type** – Controls whether to send the full PCO CHAP/PAP username or just the realm-portion in the RADIUS Access-Request message. Note that this field is only displayed if `override-pco-username` is disabled.
- **user-password-type** – Controls whether to send the full PCO CHAP/PAP password or the configured default password in the RADIUS Access-Request message.

Establishing L2TP Tunnel via RADIUS Access-Accept

The MCC can establish Layer 2 Tunneling Protocol (L2TP) tunnels based on L2TP tunnel attributes received in the RADIUS Access-Accept message.

When the MCC receives the Tunnel-Assignment-Id attribute along with other tunnel attributes in the Access-Accept message, the MCC groups subscribers into L2TP tunnels based on the Tunnel-Assignment-Id.

Subscribers with the same Tunnel-Assignment-Id are assigned to the same L2TP tunnel and a session for each subscriber is added in the corresponding tunnel. However, subscribers with the same Tunnel-Assignment-Id but different L2TP Network Server (LNS) IP addresses are assigned separate tunnels.

If the RADIUS server returns multiple tunnel attribute sets (with different tunnel tags), the MCC establishes the tunnels based on priority using the Tunnel-Preference attribute. For example, the RADIUS server can send one set of tunnel attributes for establishing an L2TP tunnel to the primary LNS and another set for establishing an L2TP tunnel to a backup LNS in case the primary LNS is not available.

If the Access-Accept message contains L2TP tunnel attributes but does not contain the Tunnel-Assignment-Id, the MCC creates an L2TP “catch-all” tunnel and adds the subscribers with no Tunnel-Assignment-Id as sessions inside this “catch-all” tunnel.

The Tunnel-Assignment-Id is reported in RADIUS Accounting-Request messages along with other L2TP tunnel attributes.

The MCC supports 400 L2TP tunnels and 7000 L2TP sessions per cluster.

Note: The MCC only processes tunnel attributes received in the RADIUS Access-Accept message when the Tunnel-Type attribute indicates L2TP.

This feature has the following limitations:

- The Tunnel-Assignment-Id attribute cannot be suppressed in Access-Accept messages and is always processed if received.
- The Tunnel-Assignment-Id cannot be configured locally.
- If the LNS with the highest priority is not available, the MCC tries to establish an L2TP tunnel with a backup LNS using additional tunnel attributes provided by the RADIUS server. However, if the L2TP tunnel associated with a subscriber goes down, the MCC does not try to establish the tunnel with the backup/additional tunnel attributes provided by the RADIUS server. In this case, the UE must tear down and re-establish the GTP tunnel.

The following commands now display the Tunnel-Assignment-Id.

```
an3000-mcm-slot7cpu1# show subscriber pdn-session <pdn-session-id>  
an3000-mcm-slot7cpu1# show data-path workflow cluster-level module l2tp  
tunnel-summary
```


Customizing the APN Reported on External Interfaces

Operators can customize the APN reported on external interfaces. The customized APN is used in the Called-Station-ID AVP and sent in messages on the following interfaces:

- Access-Request on RADIUS
- Accounting-Request on Gz and RADIUS
- Credit-Control-Request (CCR) on Gx and Gy interfaces

For Ga (GTPP) and local interfaces, the customized APN-reporting value is filled in the Access Point Name Network Identifier CDR field.

Set parameters for this feature using the following commands:

```
an3000-mcm-slot7cpu1(config)# workflow control-profile <name>
apn-reporting custom custom-reporting-apn-name <apn-name>
```

To enable this feature, set the `apn-reporting` parameter to `custom`. The value set in the `custom-reporting-apn-name` parameter is used to fill the Called-Station-Id value in messages to external interfaces.

Note: The `custom-reporting-apn-name` parameter is visible only if `apn-reporting` is set to `custom`.

Use the following command to display the reported APN name (`reporting-apn-name`) sent as the Called-Station-Id AVP on external interfaces.

```
an3000-mcm-slot7cpu1# show subscriber pdn-session <session-id>
```

Configuring APN Reporting at the Interface Level

You can configure apn-reporting at the interface level to allow for different APNs to be reported over different external interfaces. The `apn-reporting` value configured at the interface level is used to fill the Called-Station-ID AVP and User-Name AVP sent in messages on the following interfaces:

- Access-Request on RADIUS
- Accounting-Request on Gz and RADIUS
- CCR on Gx and Gy interfaces

For Ga (GTPP) and local interfaces, the `apn-reporting` value configured at the interface level is filled in the Access Point Name Network Identifier CDR field.

The `apn-reporting` value configured at the interface level overrides the `apn-reporting` value configured for the Control-Profile. If the `apn-reporting` value is not configured at the interface level, the Called-Station-ID AVP and User-Name AVP will be filled based on the `apn-reporting` value configured for the Control Profile.

```
an3000-mcm-slot7cpu1(config)# service-construct interface  
[authorization-service|accounting-service] <name> apn-reporting  
[original|mapped|prior-mapped]
```

Note: The interface-level `apn-reporting` parameter does not support the `custom` option and does not apply to the S6b interface.

Note: This feature applies to the PGW only.

Mirroring RADIUS Accounting Messages

RADIUS mirroring enables Operators to mirror RADIUS accounting messages sent on a primary RADIUS accounting interface to one or more secondary (mirrored) RADIUS accounting server location(s). With this feature, you configure two types of interfaces:

- A primary RADIUS accounting interface whose messages are mirrored
- One or more mirrored RADIUS accounting interface(s) on which the mirrored messages are sent

To configure RADIUS mirroring on a RADIUS accounting interface, perform the following steps:

- 1 Configure RADIUS-client server groups and remote RADIUS server peers for the primary RADIUS accounting interface and mirrored RADIUS accounting interface(s). See [Configuring RADIUS-Client Server Groups and Remote Peers \(page 1-9\)](#) for configuration information.
- 2 Configure the data record templates applied to RADIUS Accounting-Request (Start, Stop, Interim) messages. You can configure separate templates for each interface type (Primary/Mirrored). See [Configuring the RADIUS Accounting Service \(page 1-18\)](#) for configuration information.
- 3 Configure the RADIUS accounting event trigger template to specify the events that trigger an interim update. See [Configuring the RADIUS Accounting Service \(page 1-18\)](#) for configuration information.

Note: *The mirrored RADIUS accounting interface inherits the configuration for the event trigger template from the primary RADIUS accounting interface.*

- 4 Configure the primary RADIUS accounting interface whose messages are mirrored.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
accounting-service <primary-interface-name>
```

where **<primary-interface-name>** is the name of the primary RADIUS accounting interface.

The following table describes the configurable parameters.

Parameter	Description
interface-type	Specifies the type of RADIUS accounting interface. In this case, select primary. ■ primary – (default) Interface providing the RADIUS accounting messages to be mirrored. ■ mirrored – Interface on which the mirrored messages are sent.
is-ofc-enabled	Controls whether the system stores accounting messages locally to the disk when the remote RADIUS server is unavailable. When the remote server becomes available, the system sends the pending accounting messages. ■ true ■ false (default)
ignore-radius-failure	Controls whether the GGSN/PGW processes a RADIUS failure response or ignores the failure and continues with the Create PDP Context/Session procedure. See Configuring the ignore-radius-failure Parameter (page 1-53) for more information. ■ true – The GGSN/PGW ignores the RADIUS (authorization/accounting) failure response and continues with the Create PDP Context/Session procedure. ■ false – (default) The GGSN/PGW processes the RADIUS (authorization/accounting) failure response and rejects the Create PDP Context/Session Request. <i>Note: If the failure is a RADIUS authorization failure, MCC uses the local configuration (if configured) to create the PDP Context/Session.</i> <i>Note: Changes to this parameter take effect immediately.</i>

Parameter	Description
wait-for-accounting-response	Controls whether the GGSN/PGW waits for an Accounting-Response from the RADIUS accounting server before sending the Create/Delete PDP Context Response or Create/Delete Session Response. See Processing RADIUS Accounting-Responses (page 1-52) for more information. ■ true – (default) The GGSN/PGW waits for an Accounting-Response before sending a Create/Delete PDP Context Response or Create/Delete Session Response to the SGSN/SGW. ■ false – The GGSN/PGW sends the Create/Delete PDP Context Response or Create/Delete Session Response to the SGSN/SGW without waiting for an Accounting-Response. <i>Note: Changes to this parameter take effect immediately.</i>
pdngw Configure interim updates on the primary accounting interface: <i>Note: These parameters are only displayed when the interface type is set to primary. The mirrored accounting interface(s) inherits the configuration for these parameters from the primary accounting interface.</i>	
interim-update-interval	Specifies the maximum number of seconds that the gateway waits before generating a new accounting record.
interim-updates [enabled disabled]	Enable or disable interim updates to the RADIUS server.
data-volume-limit	Specifies the maximum number of kilobytes that the gateway maintains before generating a new accounting record.
event-trigger-template	Specifies the name of the event trigger template associated with the RADIUS accounting interface.

- 5 Associate the RADIUS-client server group and Accounting-Request data record templates with the primary RADIUS accounting interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
accounting-service <acct-service-interface-name> pdngw radius
server-group <radius-server-group-name>
[interim-template|start-template|stop-template] <template-name>
```

where

Parameter	Description
accounting-service	Specifies the primary accounting interface name.
radius server-group	Specifies the RADIUS accounting server group name. <i>Note: The RADIUS accounting server group must already exist.</i>
[interim-template start-template stop-template]	Specifies the type of data record template.

Parameter	Description
<code><template-name></code>	Specifies the name of the data record template. <i>Note: The data record templates must already exist.</i>

- 6 Configure the mirrored RADIUS accounting interface(s) on which the mirrored accounting messages are sent:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
accounting-service <mirrored-interface-name>
```

where `<mirrored-interface-name>` is the name of the mirrored RADIUS accounting interface.

Note: The mirrored RADIUS accounting interface inherits the configuration for interim updates from the primary RADIUS accounting interface.

The following table describes the configurable parameters.

Parameter	Description
<code>interface-type</code>	Specifies the type of RADIUS accounting interface. In this case, select <code>mirrored</code> . ■ <code>primary</code> – (default) Interface providing the RADIUS accounting messages to be mirrored. ■ <code>mirrored</code> – Interface on which the mirrored messages are sent.
<code>is-ofc-enabled</code>	Controls whether the system stores accounting messages locally to the disk when the remote RADIUS server is unavailable. When the remote server becomes available, the system sends the pending accounting messages. ■ <code>true</code> ■ <code>false</code> (default)

Parameter	Description
ignore-radius-failure	Controls whether the GGSN/PGW processes a RADIUS failure response or ignores the failure and continues with the Create PDP Context/Session procedure. See Configuring the ignore-radius-failure Parameter (page 1-53) for more information. ■ true – The GGSN/PGW ignores the RADIUS (authorization/accounting) failure response and continues with the Create PDP Context/Session procedure. ■ false – (default) The GGSN/PGW processes the RADIUS (authorization/accounting) failure response and rejects the Create PDP Context/Session Request. <i>Note: If the failure is a RADIUS authorization failure, MCC uses the local configuration (if configured) to create the PDP Context/Session.</i> <i>Note: Changes to this parameter take effect immediately.</i>
wait-for-accounting-response	Controls whether the GGSN/PGW waits for an Accounting-Response from the RADIUS accounting server before sending the Create/Delete PDP Context Response or Create/Delete Session Response. See Processing RADIUS Accounting-Responses (page 1-52) for more information. ■ true – (default) The GGSN/PGW waits for an Accounting-Response before sending a Create/Delete PDP Context Response or Create/Delete Session Response to the SGSN/SGW. ■ false – The GGSN/PGW sends the Create/Delete PDP Context Response or Create/Delete Session Response to the SGSN/SGW without waiting for an Accounting-Response. <i>Note: Changes to this parameter take effect immediately.</i>

- 7 Associate the RADIUS-client server group and Accounting-Request data record templates with the mirrored RADIUS accounting interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
accounting-service <mirrored-interface-name> pdngw radius server-group
<radius-server-group-name>
[interim-template|start-template|stop-template] <template-name>
```

where

Parameter	Description
accounting-interface	Specifies the mirrored RADIUS accounting interface name.
radius server-group	Specifies the RADIUS accounting server group name. <i>Note: The RADIUS accounting server group must already exist.</i>
[interim-template start-template stop-template]	Specifies the type of data record template.

Parameter	Description
<code><template-name></code>	Specifies the name of the data record template. <i>Note: The data record templates must already exist.</i>

- 8 Associate the mirrored RADIUS accounting interface with the primary RADIUS accounting interface:

```
an3000-mcm-slot7cpu1(config)# workflow control-profile
<control-profile-name> default-accounting interface
<default-interface-name> mirrored-accounting-interface-list
<mirroring-interface-name>
```

where

Parameter	Description
<code>control-profile</code>	Specifies the Workflow Control Profile name. <i>Note: The Workflow Control Profile must already exist.</i>
<code>default-accounting-interface</code>	Specifies the primary accounting interface that provides the RADIUS accounting messages to be mirrored. <i>Note: Only accounting interfaces designated with an interface type of primary can be selected.</i>
<code>mirrored-accounting-interface-list</code>	Specifies the mirroring accounting interface on which the mirrored accounting messages are sent. <i>Note: Only accounting interfaces designated with an interface type of mirrored can be selected.</i> <i>Note: This parameter is only available when a primary accounting interface is configured.</i>

Configuring DSCP Values for Control Plane Traffic

Operators can assign different levels of service to control plane traffic on RADIUS interfaces using Differentiated Services Code Point (DSCP).

RADIUS interfaces – DSCP value configured at the RADIUS server group level.

```
an3000-mcm-slot7cpu1(config)# service-construct interface  
aaa-interface-group [radius-client|radius-server] server-group <name>  
dscp-value <value>
```

Operators can also configure a DSCP value at the network-context level applicable to end marker packets, error indication messages, DNS queries and GTP-U Echo messages.

```
an3000-mcm-slot7cpu1(config)# network-context <name>  
control-dscp-value <value>
```

Note: The DSCP value configured at the RADIUS server group level overrides the network-context DSCP value.

Configuring Geo-Redundancy Health Checks on RADIUS Interfaces

Geographic Redundancy network health checks are supported on RADIUS, Gx, Gy and S6b interfaces and use both shared loopback IPs and private loopback IPs. Operators must enable health-check monitoring on at least one interface.

Note: It is recommended that you use the Geographic Redundancy Manager application in the Acuitas Service Management System to configure the Geographic Redundancy group, service endpoint, and server parameters ([step 1](#) through [step 3](#)). For instructions, see the Acuitas User's Guide.

To configure Geographic Redundancy network health checks on RADIUS interfaces:

- 1 Configure the geographic redundancy group.

```
an3000-mcm-slot8cpu1 (config) # geographic-redundancy group
<geo-redundancy-group-name> admin-state [enabled|disabled] group-id
<id> manual-state [active|standby] mode
[hot-auto|hot-manual|warm-manual] role [primary|secondary]
```

where

Parameter	Description
geographic-redundancy group	Specifies a Geographic Redundancy group name that is unique across the network.
admin-state	Specifies the administrative state: ■ enabled – Initializes and starts the redundancy mechanism (such as synchronization in hot mode). ■ disabled – Invalidates any synchronized data and sets the mechanism into a non-ready state.
group-id	Specifies the unique group ID that matches with the peer.
manual-state	Specifies the manually configured HA state (the Operator current desired state). ■ active ■ standby
mode	Specifies the instance configured HA mode. ■ warm-manual – No session synchronization occurs between nodes, failovers are Operator controlled. ■ hot-auto – Synchronization of per-session data occurs between nodes, failovers are automatically triggered based on network health monitoring results. ■ hot-manual – Synchronization of per-session data occurs.

Parameter	Description
role	Specifies the instance configured HA backup role. ■ primary – Specifies which node is “active” under normal network conditions. ■ secondary – Specifies which node is “standby” under normal network conditions.

2 Configure the service endpoint for the Geographic Redundancy server.

```
an3000-mcm-slot8cpu1(config)# network-context <network-context-name>
service-endpoint <service-endpoint-name> local-loopback-address
<ip-address> local-port-range-base <port> local-port-range-numports
<value>
```

where

Parameter	Description
network-context	Specifies a network context containing this Geographic Redundancy service endpoint.
service-endpoint	Specifies the name of the Geographic Redundancy service endpoint.
local-loopback-address	Specifies the local node address used for communication between primary and secondary nodes.
local-port-range-base	Specifies the base UDP port number used for communication between primary and secondary nodes. Must be the initial value of a contiguous block of UDP ports.
local-port-range-numports	Specifies the size of the contiguous block of ports.

3 Configure the Geographic Redundancy server parameters.

```
an3000-mcm-slot8cpu1(config)# service-construct geo-redundancy-server
<geo-redundancy-group-name> network-context <network-context-name>
service-endpoint <service-endpoint-name> remote-ip-address
<ip-address> remote-base-port <port> connectivity-monitoring-params
monitor-AAA admin-state [enabled|disabled]
```

where

Parameter	Description
geo-redundancy-server	Specifies the Geographic Redundancy group name to which this Geographic Redundancy server belongs.
network-context	Specifies the network context containing this Geographic Redundancy service endpoint.
service-endpoint	Specifies the service endpoint for this Geographic Redundancy server.
remote-ip-address	Specifies the IP address for the remote peer in the group.
remote-base-port	Specifies the port number for the remote peer in the group
connectivity-monitoring-params	Specifies network connectivity monitoring for automatic failover. Configure the parameters as follows: ■ monitor-AAA – Set to enabled to monitor the RADIUS Accounting Authentication Authorization (AAA) interface to govern fault detection and automatic failover. ■ monitor-PCRF – Set to enabled to monitor the Gx interface to govern fault detection and automatic failover. ■ monitor-OCS – Set to enabled to monitor the Gy interface to govern fault detection and automatic failover. ■ monitor-S6B – Set to enabled to monitor the S6B interface to govern fault detection and automatic failover.

- 4 Associate the Geographic Redundancy group with the RADIUS authorization and a RADIUS accounting server groups and configure health-check parameters.

Note: For each type of RADIUS server group (authorization and accounting), configure one server group with a *link-type* of *private* and another with a *link-type* of *shared*.

```
an3000-mcm-slot8cpu1(config)# service-construct interface
aaa-interface-group radius-client server-group
<radius-auth-server-group-name> geo-redundant-group
<geo-redundancy-group-name> server-group-type
[NotApplicable|Authorization|Accounting] net-health-report-enabled
true link-type [private|shared]

an3000-mcm-slot8cpu1(config)# service-construct interface
aaa-interface-group radius-client server-group
<radius-acct-server-group-name> geo-redundant-group
<geo-redundancy-group-name> server-group-type
[NotApplicable|Authorization|Accounting] net-health-report-enabled
true link-type [private|shared]
```

where

Parameter	Description
radius-client server-group	Specifies the name of the RADIUS authorization/accounting server group. The RADIUS server groups must already exist. See RADIUS Configuration (page 1-8) .
geo-redundant-group	Specifies the Geographic Redundancy group name.
server-group-type	Note: Disable the administrative state of all RADIUS server peers in the RADIUS server group before modifying this parameter. Specifies the type of RADIUS server group. ■ NotApplicable ■ Authorization ■ Accounting
net-health-report-enabled	Specifies whether to monitor network health towards the interface. To configure Geographic Redundancy network health checks on RADIUS interfaces, set to true. ■ true ■ false
link-type	Specifies whether the link used to monitor neighboring network elements is a private (local) or shared IP address among the primary and secondary network elements. ■ private ■ shared

- 5 Add peers to the RADIUS authentication and accounting server groups. See [Configuring RADIUS-Client Server Groups and Remote Peers \(page 1-9\)](#)
- 6 Configure the RADIUS authorization and accounting services, including interfaces, data-record-templates and event-triggers. See [Configuring the RADIUS Authorization Service \(page 1-15\)](#) and [Configuring the RADIUS Accounting Service \(page 1-18\)](#).
- 7 Specify the default RADIUS interfaces in the Workflow Control Profile. In this case, specify the RADIUS authorization and accounting interfaces that use a shared link.

```
an3000-mcm-slot8cpu1(config)# workflow control-profile <profile-name>  
[default-accounting-interface|default-auth-interface] <interface-name>
```

Reporting Rating-Group Level Usage Statistics

The SN-Ecs-Data-Volume VSA is used to send rating-group level usage statistics from the GGSN/PGW to the remote RADIUS server in RADIUS Accounting-Request (Start/Interim/Stop) messages.

Use the following CLI commands to configure the GGSN/PGW to report rating-group level usage statistics in the SN-Ecs-Data-Volume VSA.

- 1 Configure the Accounting-Request (Start/Stop/Interim) data record templates to include the SN-Ecs-Data-Volume VSA in Accounting-Request messages. Set the `presentFlag` parameter to `yes` to include this VSA in Accounting-Request messages. By default, the `presentFlag` parameter for the SN-Ecs-Data-Volume VSA is set to `no`.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pdngw accounting radius [start|interim|stop] <template-name>
snecsdatavolume presentFlag [conditional|no|yes]
```

- 2 Enable rating-group level usage statistics reporting on the RADIUS accounting interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
accounting-service <acct-interface-name> pdngw radius report-rg-usage
[enabled|disabled]
```

- **enabled** – The GGSN/PGW sends rating-group level statistics within the SN-Ecs-Data-Volume VSA in RADIUS Accounting-Request messages.
- **disabled** – (default) The GGSN/PGW does not send rating-group level statistics in RADIUS Accounting-Request messages.

- 3 (Optional) Configure a list of rating groups to report usage statistics for only a subset of rating groups. In this case, the SN-Ecs-Data-Volume VSA only contains statistics for the rating groups within the configured rating group quota-id range.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
accounting-service <acct-interface-name> pdngw radius
rg-quota-list-to-report <list-name> rg-quota-id-start <value>
rg-quota-id-end <value>
```

- **rg-quota-id-start** – Specifies the start of the rating group quota-id range.
- **rg-quota-id-end** – Specifies the end of the rating group quota-id range.

Note: The `rg-quota-list-to-report` parameter is only configurable when `report-rg-usage` is set to `enabled`.

Suppressing NULL Characters in PCO PAP Username/Password Fields

This feature adds a new configuration option at the APN level used to suppress trailing NULL characters in Protocol Configuration Options (PCO) PAP username/password fields before sending to the AAA server in the RADIUS Access-Request message.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>  
strip-trailing-null-pco-username-password [disabled|enabled]
```

- **disabled** – (default) The GGSN/PGW does not suppress trailing NULL characters in PCO PAP username/password fields before sending to the AAA server in the RADIUS Access-Request message.
- **enabled** – The GGSN/PGW suppresses trailing NULL characters in PCO PAP username/password fields before sending to the AAA server in the RADIUS Access-Request message.

Configuring a NAS-Identifier per APN

Use the following CLI command to configure a NAS-Identifier per APN used in RADIUS Accounting messages.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <name>
nas-identifier <string>
```

When the GGSN/PGW receives a Create PDP Context/Create Session Request message for a given APN and if the APN is configured with a NAS-Identifier, the GGSN/PGW uses the configured NAS-Identifier in RADIUS Accounting messages. If the session uses APN mapping, the mapped APN will be used to select the NAS-Identifier.

Note: When using the `nas-identifier` parameter under the APN object, do not configure the `nas-identifier` parameter under the RADIUS server group.

Note: If you do not configure the NAS-Identifier under the APN object or the RADIUS server group, the MCC gateway sends a string form of the RADIUS server peer's local loopback address in the NAS-Identifier attribute.

You can configure whether to include or exclude the NAS-Identifier attribute in the Accounting-Request (Start/Interim/Stop) data record templates associated with the RADIUS Primary interface and RADIUS Mirrored interface.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pdngw accounting radius [start|interim|stop] <template-name>
<attribute-name> presentFlag [conditional|no|yes]
```

Processing RADIUS Accounting-Responses

Configuring the wait-for-accounting-response Parameter

Using the `wait-for-accounting-response` parameter at the accounting-service level, you can control whether the GGSN/PGW waits for an Accounting-Response from the RADIUS server before sending the Create/Delete PDP Context Response or Create/Delete Session Response.

By default, the GGSN/PGW waits for an Accounting-Response from the RADIUS server before sending a Create/Delete PDP Context Response or Create/Delete Session Response to the SGSN/SGW.

Use the following command to configure the GGSN/PGW to send the Create/Delete PDP Context Response or Create/Delete Session Response to the SGSN/SGW without waiting for an Accounting-Response. The default is `true`.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
accounting-service <acct-service-interface-name>
wait-for-accounting-response false
```

Note: Changes to the `wait-for-accounting-response` parameter take effect immediately.

When using the `wait-for-accounting-response` and `ignore-radius-failure` parameters together, note the following:

wait-for-accounting-response	ignore-radius-failure	Failure Behavior
True	False	The GGSN/PGW processes the RADIUS accounting failure response and rejects the Create PDP Context/Session Request.
True	True	The GGSN/PGW ignores the RADIUS accounting failure response and sends the valid Create PDP Context/Session Response.
False	True or False	The GGSN/PGW sends the RADIUS Accounting-Request and the valid Create PDP Context/Session Response without waiting for the RADIUS Accounting-Response. The GGSN/PGW ignores the Accounting-Response when received.

Processing RADIUS Failure Responses

Configuring the ignore-radius-failure Parameter

Using the `ignore-radius-failure` parameter, you can control whether the GGSN/PGW processes a RADIUS failure response or ignores the failure and continues with the Create PDP Context/Session procedure.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
[authorization-service|accounting-service] <interface-name>
ignore-radius-failure [true|false]
```

- **true** – The GGSN/PGW ignores the RADIUS (authorization/accounting) failure response and continues with the Create PDP Context/Session procedure.
- **false** – (default) The GGSN/PGW processes the RADIUS (authorization/accounting) failure response and rejects the Create PDP Context/Session Request.

If the gateway receives an Access-Reject message within the configured time limit (`server-resp-timeout-value`), the MCC ignores the `ignore-radius-failure` setting and rejects the Create PDP Context/Session Request.

If the RADIUS authorization/accounting response is received late or not received at all within the configured time limit (RADIUS response timeout) or other internal failure occurs, the MCC will consider the `ignore-radius-failure` setting and if it is set to true, will continue with the session (by authorizing the UE locally in the case of authorization). Otherwise, the session will be discontinued.

Note: If the failure is a RADIUS authorization failure, MCC uses the local configuration (if configured) to create the PDP Context/Session.

Note: Changes to the `ignore-radius-failure` parameter take effect immediately.

Configuring the delay-failure-handling-until-all-responses Parameter

Using the `delay-failure-handling-until-all-responses` parameter, you can control whether the GGSN/PGW waits for the Accounting-Response from all of the RADIUS servers to which an Accounting-Request START message was sent before processing a RADIUS failure response. Once the GGSN/PGW receives all of the Accounting-Response messages, it sends an Accounting-Request STOP to all of the RADIUS servers, except the RADIUS servers for which the Accounting-Request START message failed.

By default, when the GGSN/PGW sends multiple RADIUS Accounting-Request START messages for a PDN session and one or more of these messages fail, the GGSN/PGW processes the failure immediately and sends Accounting-Request STOP messages to only those servers from which it had received an Accounting-Response START message and clears the session.

The `delay-failure-handling-until-all-responses` parameter can only be configured for the primary RADIUS accounting interface and will be applicable for all mirrored RADIUS accounting interfaces configured in the Control Profile, with which the primary RADIUS accounting interface is associated.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
accounting-service <acct-service-name>
delay-failure-handling-until-all-responses [true|false]
```

- **true** – The GGSN/PGW waits for the Accounting-Response from all of the RADIUS servers to which an Accounting-Request START message was sent before processing a RADIUS accounting failure.

After receiving all Accounting-Request START responses, the GGSN/PGW sends an Accounting-Request STOP message to all RADIUS servers, except the RADIUS servers for which the Accounting-Request START message failed.

- **false** – The GGSN/PGW processes the RADIUS accounting failure response immediately and sends Accounting-Request STOP messages to only those servers from which it had received an Accounting-Response START message and clears the session.

Note: Changes to the `delay-failure-handling-until-all-responses` parameter take effect immediately.

When using the `delay-failure-handling-until-all-responses` and `ignore-radius-failure` parameters together, note the following:

delay-failure-handling-until-all-responses	ignore-radius-failure	Failure from RADIUS Client Handling	Timeout at PGW Task Handling
True	False	The GGSN/PGW waits for the response from all the RADIUS servers to which an Accounting-Request START message was sent before triggering the failure action. After receiving all Accounting-Request START responses, the GGSN/PGW sends an Accounting-Request STOP message to all RADIUS servers, except the RADIUS servers for which the Accounting-Request START message failed.	The GGSN/PGW processes the RADIUS accounting failure response immediately and sends an Accounting-Request STOP message to all RADIUS servers, except the RADIUS servers for which the Accounting-Request START message failed.
True	True	The GGSN/PGW ignores the RADIUS accounting failure response and continues with the session.	The GGSN/PGW ignores the RADIUS accounting failure response and continues with the session.
False	False	The GGSN/PGW processes the RADIUS accounting failure response immediately and sends Accounting-Request STOP messages to only those servers from which it had received an Accounting-Response START message and clears the session.	The GGSN/PGW processes the RADIUS accounting failure response immediately and sends Accounting-Request STOP messages to only those servers from which it had received an Accounting-Response START message and clears the session.
False	True	The GGSN/PGW ignores the RADIUS accounting failure response and continues with the session.	The GGSN/PGW ignores the RADIUS accounting failure response and continues with the session.

Processing the TMO-Virtual-Gi-ID VSA

Use the following command to specify whether to process the TMO-Virtual-Gi-ID VSA when received in the Access-Accept message from the RADIUS server.

```
an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name>
vsa-virtual-gi-id-enabled [true|false] vsa-virtual-gi-id
<default-value>
```

where

Parameter	Description
vsa-virtual-gi-id-enabled	Specifies whether to process the TMO-Virtual-Gi-ID VSA when received in the Access-Accept message from the RADIUS server. ■ true – If the MCC gateway receives the TMO-Virtual-Gi-ID VSA in the Access-Accept message, it sends the VSA in Accounting-Request messages (Start/Interim/Stop) on the RADIUS interface and in CCR-I messages on the Gx and Gy interfaces. ■ false – (default) If the MCC gateway receives the TMO-Virtual-Gi-ID VSA in the Access-Accept message, it does not send the VSA in Accounting-Request messages on the RADIUS interface or in CCR-I messages on the Gx and Gy interfaces.
vsa-virtual-gi-id	Specifies a default value for the TMO-Virtual-Gi-ID VSA. If the vsa-virtual-gi-id-enabled parameter is set to true but the MCC gateway does not receive the TMO-Virtual-Gi-ID VSA in the RADIUS Access-Accept message, the MCC gateway uses this default value to fill the TMO-Virtual-Gi-ID VSA sent in RADIUS Accounting-Request messages (Start/Interim/Stop) and in CCR-I messages on the Gx and Gy interfaces. <i>Note: This parameter is only visible in the CLI when the vsa-virtual-gi-id-enabled parameter is set to true.</i>

You can configure whether to include or exclude the TMO-Virtual-Gi-ID VSA in RADIUS Access-Accept and Accounting-Request (Start/Stop/Interim) messages and Gx/Gy CCR-I messages.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pgw auth radius access-accept <template-name> tmo_virtual_gi_id
presentFlag [conditional|no|yes]
```

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pdngw accounting radius [start|interim|stop] <template-name>
tmo_virtual_gi_id presentFlag [conditional|no|yes]
```

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pgw [pcrf|ocs] diameter start <template-name> tmovirtualgiid
presentFlag [yes|no|conditional]
```

Monitoring

This section describes how to monitor the operational status of the remote RADIUS server and how to display statistical information for RADIUS communication messages. This section also describes the alarm notification that occurs when the remote RADIUS server goes down. For more information on statistics, see *Acuitas Performance Statistics*.

RADIUS-Client Server Group Status

Use the following CLI command to display the operational status of a specific RADIUS-client server group.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group radius-client server-status server-group-status
<radius-server-group-name>
```

The system displays output similar to the following:

```
OPERATIONAL
NAME          STATUS
-----
AAAGROUP1 up
```

Remote RADIUS Server Peer Status

Use the following CLI command to display the operational status of a specific remote RADIUS server peer.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group radius-client server-status
server-group-peer-status <radius-server-group-name> <radius-peer-name>
```

The system displays output similar to the following:

```
RADIUS
SERVER
GROUP      RADIUS PEER  OPERATIONAL  ADMIN
NAME       NAME          STATUS       STATUS      PRIORITY
-----
AAAGROUP1  RADIUS-AUTH1  up           enabled 1
```

Note: The Priority number does not display for peers with *peer-retry-mode* set to either *Sequential* or *Alternate*.

Detailed Remote RADIUS Server Peer Status

Use the following CLI command to display detailed status information for a specific remote RADIUS server peer.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group radius-client server-status peer-status
<radius-server-group-name> <radius-peer-name> <slot>
```

The system displays output similar to the following:

```
operational-status down
operational-reason "Radius Server Timed Out"
admin-status enabled
local-ip-address 10.53.131.27
local-port-start 6001
local-port-end 6004
remote-ip-address 10.48.148.2
remote-port 9271
```

RADIUS Statistics

You can display statistical information for RADIUS communication messages for the grid, cluster, and slot level. For each level, you can also display statistics for all remote RADIUS server peers or for a specific remote RADIUS server peer as shown in the following examples. For more information on statistics, see *Acuitas Performance Statistics*.

RADIUS Statistics at the Grid Level

Use the following CLI command to display the RADIUS service procedure grid statistics for all peers.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group radius-client statistics grid-level proc-stats all
<grid>
```

Use the following CLI command to display the RADIUS service procedure grid statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group radius-client statistics grid-level proc-stats
peer <peer-name>
```

RADIUS Statistics at the Cluster Level

Use the following CLI command to display the RADIUS service procedure cluster statistics for all peers.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group radius-client statistics cluster-level proc-stats
all <cluster-id>
```


Use the following CLI command to display the RADIUS service procedure cluster statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group radius-client statistics cluster-level proc-stats
peer <cluster-id> <peer-name>
```

RADIUS Statistics at the Slot Level

Use the following CLI command to display the RADIUS service procedure slot statistics for all peers.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group radius-client statistics slot-level proc-stats all
<cluster-id> <slot>
```

Use the following CLI command to display the RADIUS service procedure slot statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group radius-client statistics slot-level proc-stats
peer <cluster-id> <slot> <peer-name>
```

Data Usage Cumulative Statistics

Use the following CLI commands to display cumulative statistics for data usage at the interface level.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
data-usage-stats interface-usage-stats <gw-type> <gw-name>
<interface-type> <interface-name>
```

- totalUplinkBytes/totalDownlinkBytes
- totalUplinkPackets/totalDownlinkPackets

Alarms

A major-level trap similar to the following occurs when a remote RADIUS server peer goes down. The alarm clears after the connection is restored.

anRadiusPeerDown

Description:	The RADIUS peer health check failed and this RADIUS peer is declared down.
Trap ID:	91
Status:	Current
Severity:	Major
Auto Clearing:	Yes
Potential Impact:	If this RADIUS peer is configured for authentication and there is no alternate operationally Up RADIUS peer, then the new session setups fail. If this RADIUS peer is used for accounting and if there is no alternate operationally Up RADIUS peer and if is-ofc-enabled is enabled on Accounting Service, then the accounting records are stored in the AN3000 and auto played after one of the RADIUS peers recovers.
Corrective Action:	Check the routing/connectivity towards the RADIUS peer. The alarm clears after connectivity is restored.

Failover Handling

This section describes the RADIUS failover handling supported on the system and provides message flows illustrating different RADIUS failure scenarios:

- [RADIUS Server Selection During Failover \(page 1-62\)](#)
- [RADIUS Health Check \(page 1-64\)](#)
- [Enhanced RADIUS Health Check \(page 1-66\)](#)
- [Failover Message Flows \(page 1-68\)](#)

Note: For failover information when using the Geographic Redundancy Manager, see the *Acuitas User's Guide*.

RADIUS Server Selection During Failover

If a RADIUS server peer within the same RADIUS server group is unavailable, the RADIUS client (running on the MCC gateway) tries to connect with another peer within the group. MCC supports the following methods for selecting a RADIUS server peer during a failover.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group radius-client server-group
<radius-server-group-name> peer-retry-mode
[Sequential|Alternate|alternate-with-priority]
```

- **Sequential** – If the MCC gateway does not receive a response from a remote RADIUS server peer within the configured time limit (`server-resp-timeout-value`), it retransmits the request to the *same* peer (until the request retry limit, `max-num-response-timeout`, is reached) before selecting another active peer in the same group in a round-robin fashion.
- **Alternate** – (default) If the MCC gateway does not receive a response from a remote RADIUS server peer within the configured time limit (`server-resp-timeout-value`), it retransmits the request to the next active peer in the same group in a round-robin fashion. *When this mode is selected, the configured priority does not apply.*
- **alternate-with-priority** – All new requests are sent to the RADIUS server peer with the highest priority (with 1 being the highest). If the RADIUS server peer with the highest priority is unavailable, the request is sent to the peer with the next highest priority in the same group. If multiple RADIUS server peers are assigned the highest priority (1), the RADIUS client uses a round-robin approach to select from among those peers for new requests.

Note: For information on how to configure the RADIUS server selection method, see [Configuring RADIUS-Client Server Groups and Remote Peers \(page 1-9\)](#).

The following table describes the RADIUS failover parameters, configurable at the peer-level.

Parameter	Description
<code>max-num-response-timeout</code>	The maximum number of retransmission attempts allowed for a server group after which the RADIUS client (running on the MCC gateway) rejects the request. Enter a value between 1-5 (default is 3). <i>Note: Affirmed Networks recommends that all RADIUS server peers within a server group use the same value for this parameter. If the peers use different values, any of the values may apply.</i>
<code>server-resp-timeout-value</code>	The number of seconds to wait before the system retransmits the request. Enter a value between 1-5 seconds (default is 1 second).

RADIUS Health Check

The system uses the RADIUS Health Check feature to detect a non-responsive remote RADIUS server peer by using a counter to track the number of consecutive unanswered requests (`server-dead-num-attempts`) for a given peer. If the number of consecutive unanswered requests is reached on a particular peer, the system:

- Marks the peer as down.
- Generates a TRAP.
- Starts the dead server declare timer (`server-dead-timeout-value`).

When the dead server declare timer expires (`server-dead-timeout-value`), the RADIUS client (running on the MCC gateway) verifies the peers availability by sending a single RADIUS request (Access/Accounting) to the affected peer and waiting for a valid response. During this time, the peer's state is pending in that the system does not use the affected peer for any other transactions.

- If the peer returns a valid response, the system changes the peer's operational state to active and restores the peer to full service.
- If the peer does not respond with a valid response within the configured time limit (`server-resp-timeout-value`), the system transmits the RADIUS request to another active peer from the RADIUS server group and moves the non-responsive peer from a pending state to an inactive state. The verification cycle begins again, except the system does not generate a TRAP.

Note: The system displays the operational status of the affected RADIUS server peer as down until the system receives a valid response for the request used to verify the peer's availability.

Configure the RADIUS Health Check parameters at the peer level using the following CLI command:

```
an3000-mcm-slot7cpu1(config)# service-construct interface  
aaa-interface-group radius-client server-group  
<radius-server-group-name> peer <radius-peer-name>
```

The following table describes the configurable RADIUS Health Check parameters.

Parameter	Description
<code>server-dead-num-attempts</code>	The number of consecutive unanswered requests allowed for a peer before the system declares a RADIUS server peer down. <i>Note: Setting this parameter to a value lower than 500 may result in the RADIUS server peer frequently being taken out of service at very high call rates.</i>
<code>server-dead-timeout-value</code>	The number of seconds to wait before attempting to reconnect to the non-responsive RADIUS server peer after moving the peer from a pending state to an inactive state.

Enhanced RADIUS Health Check

Enhanced RADIUS Health Check (ERHC) ensures the remote RADIUS server peer is operational before declaring it active by periodically sending RADIUS Authorization/Accounting Health Check messages to RADIUS server peers to check whether they are responding.

Note: You can configure the time interval between remote RADIUS Authentication/Accounting Health Check messages using the `remote-health-check-period` parameter. This timer continuously runs in the background and starts automatically as soon as it expires.

When you enable ERHC, the system periodically sends a RADIUS Authorization/Accounting Health Check message (depending on the type of server group) to a RADIUS server peer and waits for a response before declaring it operational.

- If the remote peer sends a valid response, the RADIUS client declares the peer operational.
- If the remote peer does not respond within the configured ERHC response time limit (`remote-health-check-response-period`), the RADIUS client retransmits the RADIUS Authorization/Accounting Health Check message and starts the ERHC response timer (`remote-health-check-response-period` timer).

When the timer expires, the RADIUS client again sends a RADIUS Authorization/Accounting Health Check message to the peer and waits for a response. This cycle continues until the RADIUS client receives a valid response from the remote peer or the ERHC retry limit is reached (`remote-health-check-retry-count`).

If the ERHC retry limit (`remote-health-check-retry-count`) is reached, the RADIUS client marks the RADIUS server peer as down and repeats the ERHC process.

Configure ERHC parameters at the server-group level using the following CLI command:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group radius-client server-group
<radius-server-group-name>
```

The following table describes the configurable Enhanced RADIUS Health Check parameters.

Field	Description
remote-health-check	Enable or disable ERHC used to monitor the status of remote RADIUS server peers within a RADIUS server group. ■ true – ERHC is enabled. ■ false – (default) ERHC is disabled.
remote-health-check-period	Number of seconds between remote RADIUS Authentication/Accounting Health Check messages. The default is 15. The range is 1-300 seconds.
remote-health-check-response-period	Number of seconds the RADIUS client waits for a response after sending a RADIUS Authentication/Accounting Health Check message to the remote RADIUS server peer. The default is 1. The range is 1-10 seconds.
remote-health-check-retry-count	Maximum number of health check attempts to a RADIUS server peer before declaring the peer down. The default is 3. The range is 0-10.

Failover Message Flows

This section illustrates the following RADIUS failover message flows:

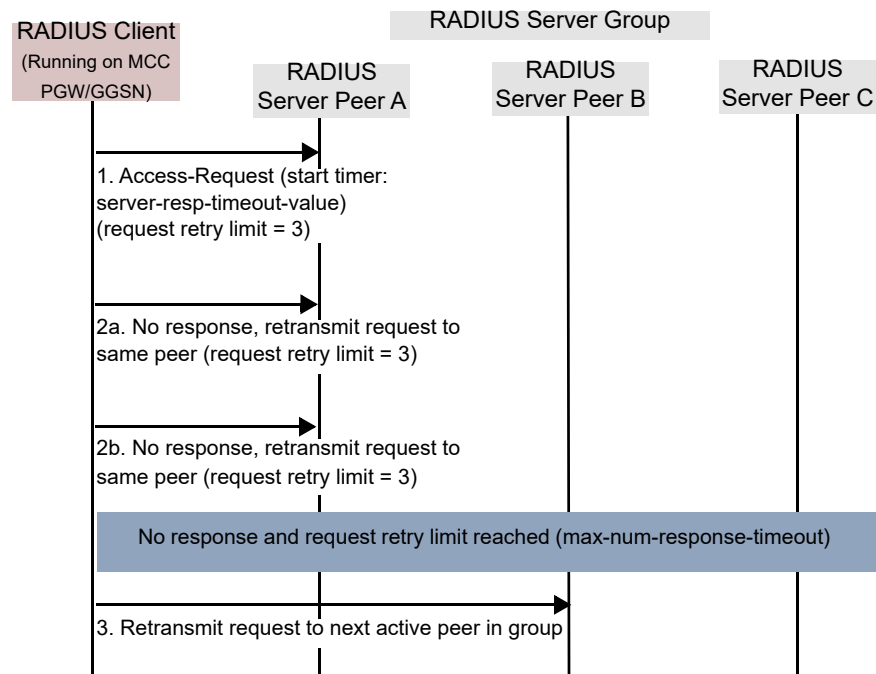
- [Sequential RADIUS Server Peer Selection \(page 1-69\)](#)
- [Alternate RADIUS Server Peer Selection \(page 1-70\)](#)
- [Alternate-With-Priority RADIUS Server Peer Selection \(page 1-71\)](#)
- [RADIUS Health Check Message Flow \(page 1-72\)](#)
- [Enhanced RADIUS Health Check Message Flow \(page 1-73\)](#)

Note: *If all the RADIUS server peers within a RADIUS server group are down, the RADIUS client rejects the request.*

Sequential RADIUS Server Peer Selection

Figure 1-2 illustrates the message flow for failover handling using the Sequential RADIUS server peer selection.

Figure 1-2 Failover: Sequential RADIUS Server Selection

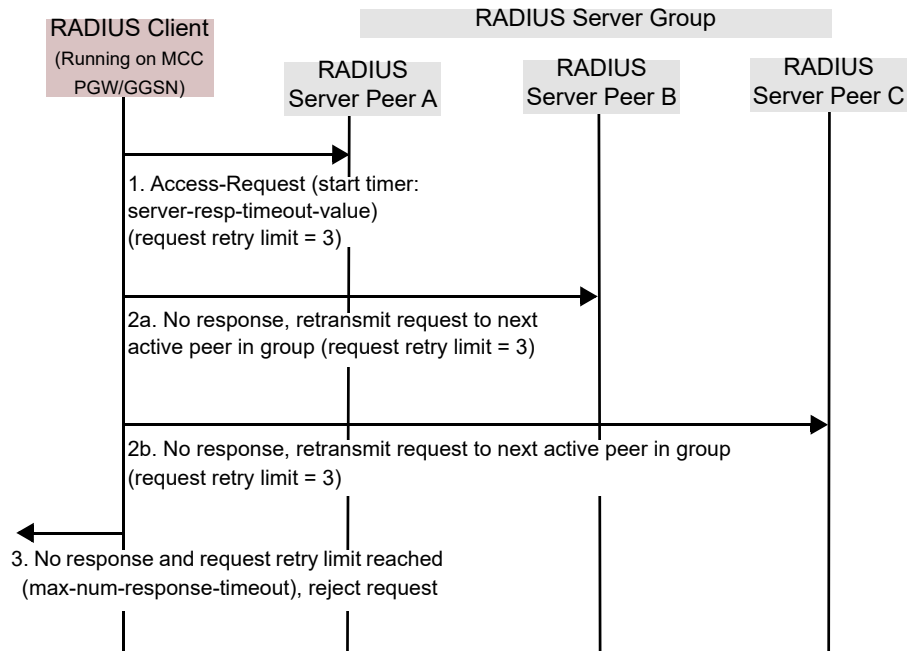


- 1 When you add a RADIUS server peer to a RADIUS-client server group, the system places the peer on the active list and sends an Access-Request message to the first available RADIUS server peer in the active list. This message contains the username, encrypted password and shared key.
- 2 If the RADIUS server peer does not respond within the configured time limit (`server-resp-timeout-value`):
 - a The RADIUS client retransmits the request to the same peer.
 - b This cycle continues until the RADIUS client receives a response or the configured request retry limit (`max-num-response-timeout`) is reached.
- 3 If Peer A does not respond within the time limit (`server-resp-timeout-value`) and the configured request retry limit (`max-num-response-timeout`) is reached, the RADIUS client retransmits the request to the next active peer in the group.

Alternate RADIUS Server Peer Selection

Figure 1-3 illustrates the message flow for failover handling using the Alternate RADIUS server peer selection.

Figure 1-3 Failover: Alternate RADIUS Server Peer Selection



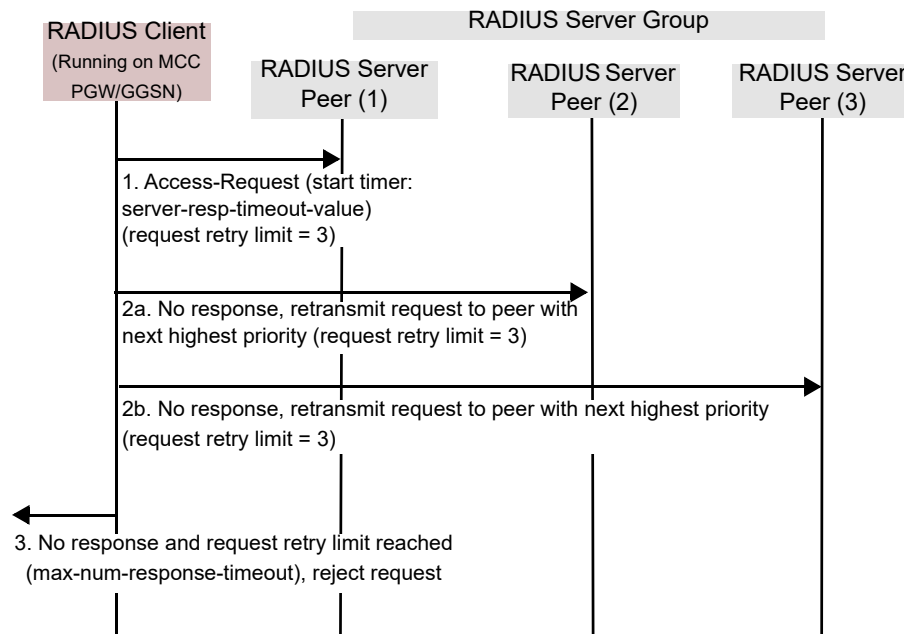
- 1 When you add a RADIUS server peer to a RADIUS-client server group, the system places the peer on the active list and sends an Access-Request message to the first available RADIUS server peer in the active list. This message contains the username, encrypted password and shared key.
- 2 If the first available RADIUS server peer (Peer A) does not respond within the configured time limit (`server-resp-timeout-value`):
 - a The RADIUS client retransmits the request to the next active peer in the server group (Peer B).
 - b This cycle continues until the RADIUS client receives a response or the configured request retry limit (`max-num-response-timeout`) is reached.
- 3 If Peer C does not respond within the time limit (`server-resp-timeout-value`) and the configured request retry limit (`max-num-response-timeout`) is reached, the RADIUS client rejects the request.

Note: In this example, the maximum number of retransmission attempts allowed for this server group has been reached based on the request retry limit (`max-num-response-timeout`) configured for the most recently used peer, in this case, Peer C. If the request retry limit on Peer C were set to 4, the system would retransmit the request once again to the next active peer in the list.

Alternate-With-Priority RADIUS Server Peer Selection

Figure 1-4 illustrates the message flow for failover handling using the Alternate-With-Priority RADIUS server peer selection.

Figure 1-4 Failover: Alternate-With-Priority RADIUS Server Peer Selection



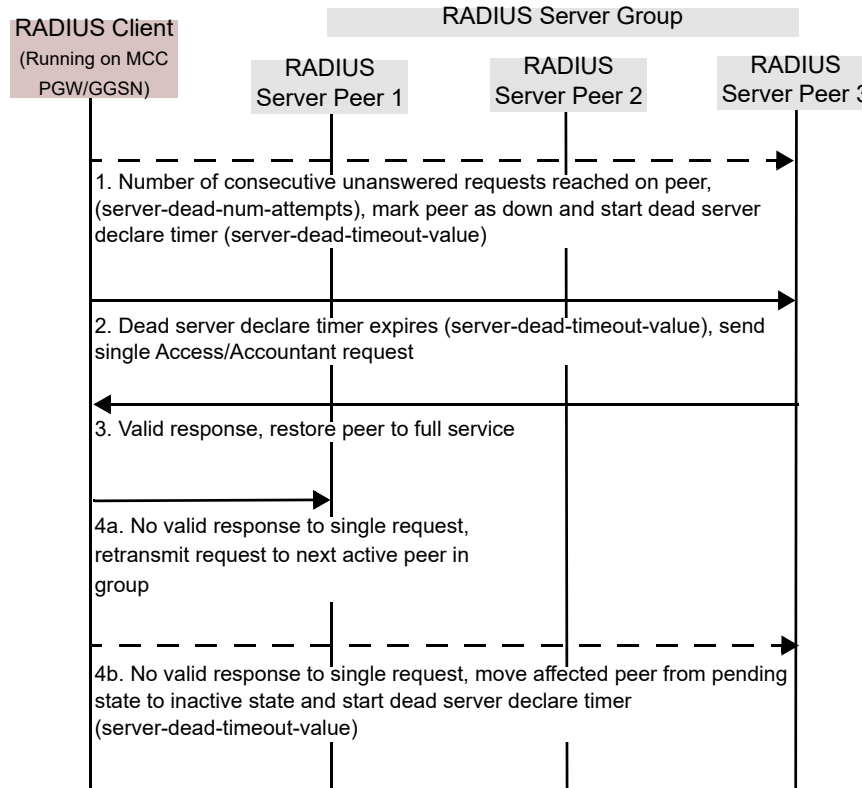
- 1 When you add a RADIUS server peer to a RADIUS-client server group, you assign the peer a priority value (with 1 being the highest priority). The system places the peer on the active list and sends an Access-Request message to the RADIUS server peer with the highest priority. This message contains the username, encrypted password and shared key.
- 2 If the RADIUS server peer with the highest priority (1) does not respond within the configured time limit (`server-resp-timeout-value`):
 - a The RADIUS client retransmits the request to the peer with the next highest priority in the server group.
 - b This cycle continues until the RADIUS client receives a response or the configured request retry limit (`max-num-response-timeout`) is reached.
- 3 If the RADIUS server peer (with priority 3) does not respond within the time limit (`server-resp-timeout-value`) and the configured request retry limit (`max-num-response-timeout`) is reached, the RADIUS client rejects the request.

Note: If multiple RADIUS server peers are assigned the highest priority (1), the RADIUS client uses a round-robin approach to select from among those peers for new requests.

RADIUS Health Check Message Flow

Figure 1-5 illustrates the message flow for RADIUS Health Check failure detection.

Figure 1-5 RADIUS Health Check Message Flow

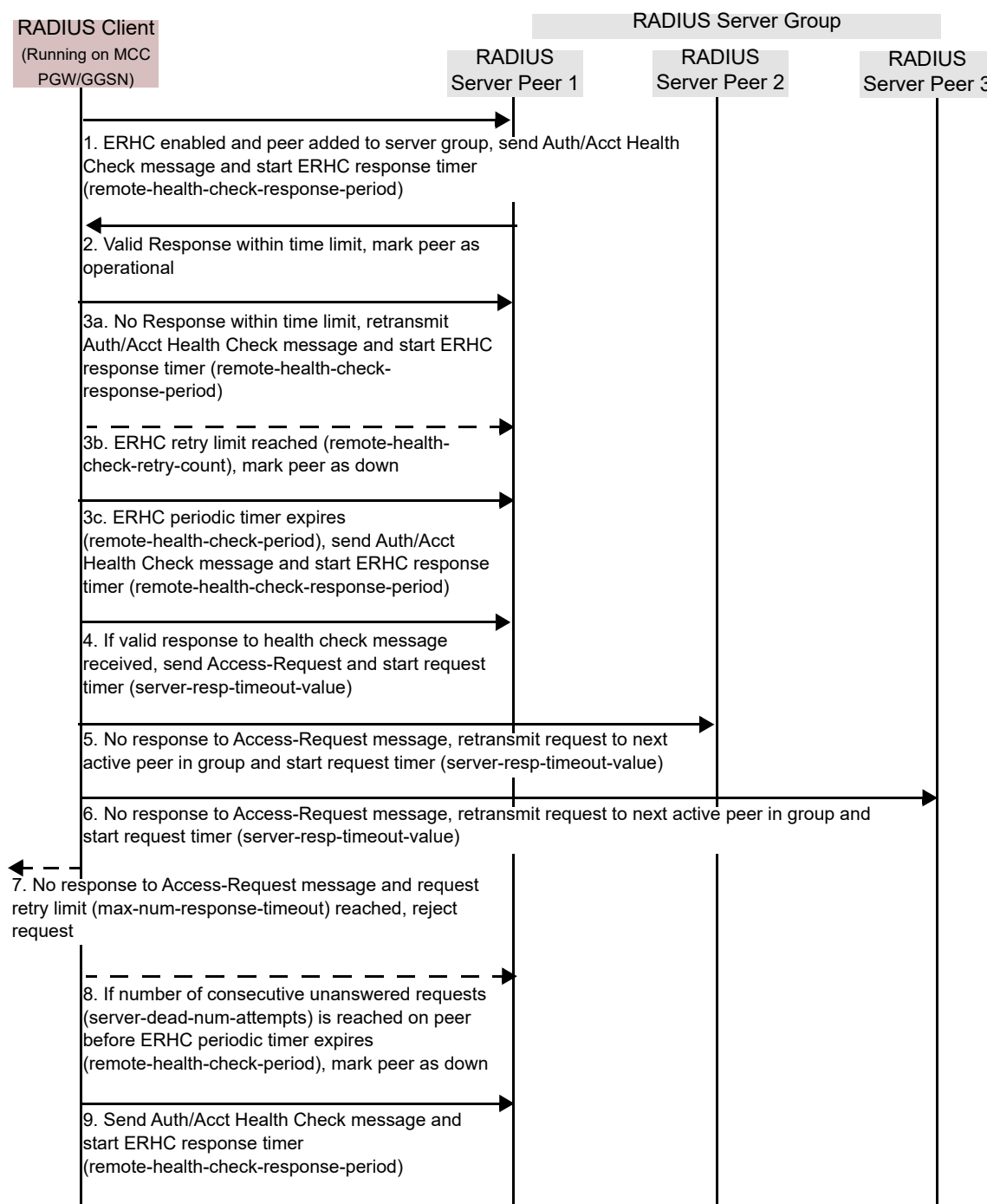


- 1 If the number of consecutive unanswered requests (`server-dead-num-attempts`) is reached on a particular peer, the system:
 - Marks the peer as down.
 - Generates a TRAP.
 - Starts the dead server declare timer (`server-dead-timeout-value`).
- 2 When the dead server declare timer expires, the RADIUS client verifies the peers availability by sending a single RADIUS request (Access/Accounting) to the affected peer and waiting for a valid response. During this time, the peer's state is pending in that the system does not use the affected peer for any other transactions.
- 3 If the peer returns a valid response, the system changes the peer's operational state to active and restores the peer to full service.
- 4 If the peer does not respond with a valid response within the configured time limit (`server-resp-timeout-value`), the system:
 - a Selects another active peer from the RADIUS server group and retransmits the request.
 - b Moves the unresponsive peer from a pending state to an inactive state and the cycle begins again, except the system does not generate a TRAP.

Enhanced RADIUS Health Check Message Flow

Figure 1-6 illustrates the message flow for Enhanced RADIUS Health Check failure detection.

Figure 1-6 Enhanced RADIUS Health Check Message Flow



- 1 When ERHC is enabled and you add a RADIUS server peer to a server group, the system sends a RADIUS Authorization/Accounting Health Check message (depending on the type of server group) to the RADIUS server peer and starts the ERHC response timer (`remote-health-check-response-period`).

***Note:** When ERHC is enabled, the system periodically sends RADIUS Authorization/Accounting Health Check messages to the RADIUS server peers within a server group to check whether they are responding. You can configure the number of seconds between remote RADIUS Authentication/Accounting Health Check messages using the `remote-health-check-period` parameter. This timer continuously runs in the background and starts automatically as soon as it expires.*

- 2 If the system receives a valid response from the remote peer within the configured time limit (`remote-health-check-response-period`), the RADIUS client marks the peer as operational and sends RADIUS requests to the peer.
- 3 If the system does not receive a response from the remote peer within the configured time limit (`remote-health-check-response-period`), the following can occur:
 - a The RADIUS client retransmits the RADIUS Authorization/Accounting Health Check message and starts the ERHC response timer (`remote-health-check-response-period`). This cycle continues until the RADIUS client receives a valid response from the remote peer or the ERHC retry limit is reached (`remote-health-check-retry-count`).
 - b If the ERHC retry limit (`remote-health-check-retry-count`) is reached, the RADIUS client marks the RADIUS server peer as down.
 - c When the ERHC periodic timer expires (`remote-health-check-period`), the RADIUS client again sends a RADIUS Authorization/Accounting Health Check message to the RADIUS server peer and starts the ERHC response timer (`remote-health-check-response-period`). This cycle continues until the RADIUS client receives a valid response from the RADIUS server peer.
- 4 The RADIUS client sends an Access-Request to the RADIUS server peer and starts the response timer (`server-resp-timeout-value`).
- 5 If the first available RADIUS server peer (Peer 1) does not respond within the configured time limit (`server-resp-timeout-value`), the RADIUS client retransmits the request to the next active peer in the server group (Peer 2).
- 6 If Peer 2 does not respond within the time limit, the RADIUS client retransmits the request to the next active peer in the server group (Peer 3). This cycle continues until the RADIUS client receives a valid response from the remote peer or the configured request retry limit (`max-num-response-timeout`) is reached.
- 7 If the next available peer in the RADIUS server group does not respond within the time limit (`server-resp-timeout-value`) and the configured request retry limit (`max-num-response-timeout`) is reached for the server group, the RADIUS client rejects the request.

- 8 If the number of consecutive unanswered requests (`server-dead-num-attempts`) is reached on a particular peer before the ERHC periodic timer expires (`remote-health-check-period`), the system marks the peer as down.
- 9 The system immediately sends a RADIUS Authorization/Accounting Health Check message to the RADIUS server peer and starts the ERHC response timer (`remote-health-check-response-period`). This cycle continues until the RADIUS client receives a valid response from the RADIUS server peer.

Note: *If all the RADIUS server peers within a RADIUS server group are down, the RADIUS client rejects the request.*

Message Definitions

Table 1-1 describes the message definitions for the RADIUS interfaces.

Table 1-1 RADIUS Interface Message Definitions

Message Type	Decimal Code	Reference
Access-Request (page 1-77)	1	RFC 2865
Access-Accept (page 1-81)	2	RFC 2865
Access-Reject (page 1-83)	3	RFC 2865
Accounting-Request (page 1-84)	4	RFC 2866
Accounting-Response (page 1-93)	5	RFC 2866
Disconnect-Request (page 1-94)	40	RFC 5176
Disconnect-ACK (page 1-96)	41	RFC 5176
Disconnect-NAK (page 1-96)	42	RFC 5176

Access-Request

The GGSN/PGW sends an Access-Request message to the RADIUS server requesting authentication and access to specific network services on behalf of a particular user.

Table 1-2 lists the Access-Request message attributes. For a description of each attribute, see [Chapter 2, RADIUS AVP Descriptions](#).

Table 1-2 Access-Request Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
User-Name	1	String	Mandatory
User-Password	2	String	Conditional (Present if PAP is used)
CHAP-Password	3	String	Conditional (Present if CHAP is used)
NAS-IP-Address	4	IPv4	Conditional (NAS-IP-Address or NAS-Identifier must be present)
Service-Type	6	2 (Framed) or 17 (Authorize Only)	Optional
Framed-Protocol	7	7 (GPRS PDP Context)	Optional
Framed-IP-Address	8	IPv4	Conditional
Called-Station-Id	30	APN (UTF-8 encoded characters)	Mandatory
Calling-Station-Id	31	MSISDN in international format (UTF8 encoded decimal)	Optional
NAS-Identifier	32	String	Conditional (NAS-IP-Address or NAS-Identifier must be present)

Table 1-2 Access-Request Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
Acct-Session-Id	44	GGSN/PGW IP Address (IPv4 or IPv6) and Charging-ID concatenated in a UTF-8 encoded hexadecimal character	Conditional
Acct-Multi-Session-Id	50	String	Optional
Event-Timestamp	55	64 Bit Unsigned Integer	Optional
CHAP-Challenge	60	String	Conditional (Present if CHAP is used)
NAS-Port-Type	61	As per RFC 2865	Optional
NAS-Port-Id	87	String	Optional
Chargeable-Unit-Identity	89	String	Optional
NAS-IPv6-Address	95	IPv6	Conditional (NAS-IPv6-Address or NAS-Identifier must be present)
Framed-IPv6-Prefix	97	IPv6	Conditional
3GPP Vendor-Specific	26/10415	As per section 16.4.7.2 of 3GPP TS 29.061 V11.0.0 (2012-03)	
3GPP-IMSI	1	String	Optional
3GPP-Charging-Id	2	Integer	Optional
3GPP-PDP-Type	3	Integer	Conditional (Mandatory if Framed-Protocol is present)
3GPP-CG-Address	4	IPv4	Optional
3GPP-GPRS-Negotiated-QoS-Profile	5	String	Optional
3GPP-SGSN-Address	6	IPv4	Optional
3GPP-GGSN-Address	7	IPv4	Optional

Table 1-2 Access-Request Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
3GPP-IMSI-MCC-MNC	8	String	Optional
3GPP-GGSN-MCC-MNC	9	String	Optional
3GPP-NSAPI	10	String	Optional
3GPP-Selection-Mode	12	String	Optional
3GPP-Charging-Characteristics	13	String	Optional
3GPP-CG-IPv6-Address	14	IPv6	Optional
3GPP-SGSN-IPv6-Address	15	IPv6	Optional
3GPP-GGSN-IPv6-Address	16	IPv6	Optional
3GPP-IPv6-DNS-Servers	17	IPv6	Optional
3GPP-SGSN-MCC-MNC	18	String	Optional
3GPP-Teardown-Indicator	19	String	Optional
3GPP-IMEISV	20	String	Optional
3GPP-RAT-Type	21	String	Optional
3GPP-User-Location-Info	22	String	Optional
3GPP-MS-TimeZone	23	String	Optional
3GPP-CAMEL-Charging-Info	24	String	Optional
3GPP-Packet-Filter	25	String	Optional
3GPP-Negotiated-DSCP	26	String	Optional
3GPP-Allocate-IP-Type	27	String	Conditional

[Table 1-3](#) lists the vendor-specific attributes (VSAs) for Access-Request messages. For a description of each attribute, see [Chapter 2, RADIUS AVP Descriptions](#).

Table 1-3 Vendor-Specific Access-Request Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
3GPP2-GMT-Time-Zone-Offset	26/5535/143	Signed Integer	Optional
3GPP2-Carrier-ID	26/5535/142	String	Optional
SN1-Service-Type	26/8164/24	Integer	Optional
SN1-GTP-Version	26/8164/62	Integer	Optional
AF-Dhcp-Option82-Id	26/37963/25	String	Optional
AF-Dhcp-Option82-Circuit-Id	26/37963/27	String	Optional
AF-Dhcp-Option82-Remote-Id	26/37963/28	String	Optional

Access-Accept

The RADIUS server sends an Access-Accept message in response to an Access-Request message to indicate that the user is authenticated and authorized to access the requested network services. The Access-Accept message also contains the configuration information necessary to deliver service to the user.

[Table 1-4](#) lists the Access-Accept message attributes. For a description of each attribute, see [Chapter 2, RADIUS AVP Descriptions](#).

Table 1-4 Access-Accept Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
User-Name	1	String	Optional
Service-Type	6	Framed	Optional
Framed-Protocol	7	7 (GPRS PDP Context)	Optional
Framed-IP-Address	8	IPv4	Conditional
Framed-IP-Netmask	9	IPv4	Conditional
Framed-MTU	12	String	Optional
Framed-Route	22	String As Per IETF RFC 2865 Section 5.22	Optional
Class	25	String	Optional
Session-Timeout	27	32 Bit Unsigned Integer	Optional
Idle-Timeout	28	32 Bit Unsigned Integer	Optional
Called-Station-Id	30	APN (UTF-8 encoded characters)	Optional
Tunnel-Type	64	32 Bit Unsigned Integer	Optional
Tunnel-Medium-Type	65	32 Bit Unsigned Integer	Optional
Tunnel-Client-Endpoint	66	String	Optional
Tunnel-Server-Endpoint	67	String	Optional
Tunnel-Password	69	String	Optional
Tunnel-Assignment-ID	82	String	Optional

Table 1-4 Access-Accept Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
Tunnel-Preference	83	32 Bit Unsigned Integer	Optional
Framed-Pool	88	String	Optional
Tunnel-Client-Auth-ID	90	String	Optional
Tunnel-Server-Auth-ID	91	String	Optional
Framed-Interface-ID	96	64 Bits As Per IETF RFC 3162 [50]	Optional
Framed-IPv6-Pool	97	IPv6	Conditional
Framed-IPv6-Route	99	String As Per IETF RFC 3162 Section 2.5	Optional
Framed-IPv6-Pool	100	String	Optional
Delegated-IPv6-Prefix	123	IPv6	Conditional

[Table 1-5](#) lists the VSAs for Access-Accept messages. For a description of each attribute, see [Chapter 2, RADIUS AVP Descriptions](#).

***Note:** The MCC ignores unknown VSAs received in RADIUS Access-Accept messages and continues with the session setup. The MCC will only generate parsing errors for known VSAs.*

Table 1-5 Vendor-Specific Access-Accept Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
Nokia-Session-Charging-Type	26/94/11	String	Optional
Nokia-OCS-ID1	26/94/12	Integer	Optional
Nokia-OCS-ID2	26/94/13	Integer	Optional
Nokia-Requested-APN	26/94/15	String	Optional
AF-Rule-Base-Name	26/37963/1	String	Optional
AF-Rule-Name	26/37963/2	String	Optional
AF-IP-POOL-Name	26/37963/3	String	Optional

Table 1-5 Vendor-Specific Access-Accept Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
AF-BwFairApp-Name	26/37963/4	String	Optional
AF-Charging-Instance-Name	26/37963/7	String	Optional
AF-Default-QoS-Profile-Name	26/37963/6	String	Optional
AF-SLA-Profile-ID	26/37963/21	String	Optional
AF-Advice-of-Charge	26/37963/22	Integer	Optional
AF-Workflow-Control-Profile-Name	26/37963/23	String	Optional
AF-Workflow-Data-Profile-Name	26/37963/24	String	Optional
AF-Msisdn	26/37963/26	String	Optional
AF-EDR-Interface-Name	26/37963/29	String	Optional
MS-Primary-DNS-Server	26/311/28	IPv4	Optional
MS-Secondary-DNS-Server	26/311/29	IPv4	Optional
MS-Primary-NBNS-Server	26/311/30	IPv4	Optional
MS-Secondary-NBNS-Server	26/311/31	IPv4	Optional
3GPP-Charging-Characteristics	26/10415/13	String	Optional
3GPP-SGSN-MCC-MNC	26/10415/18	String	Optional
Cisco-AV-Pair	26/9/1	String	Optional
TMO-Charging-Rule-Base-Name	26/29168/1	String	Optional
TMO-Transparent-Data	26/29168/2	Octet String	Optional
TMO-Virtual-Gi-ID	26/29168/15	32 Bit Integer	Optional

Access-Reject

The RADIUS server sends the GGSN/PGW an Access-Reject message in response to an Accept-Request message to indicate that the user is not authorized to access the requested network services.

Accounting-Request

The GGSN/PGW sends an Accounting-Request message to the RADIUS server to convey information about the authorized network service. Accounting-Requests contain the following message types:

- START – Starts a user session.
- STOP – Ends a user session.
- INTERIM – Updates a user session.
- ON – Indicates that the MCC device is up.
- OFF – Indicates that the MCC device is down.

Accounting-Request START

[Table 1-6](#) lists the Accounting-Request START message attributes. For a description of each attribute, see [Chapter 2, RADIUS AVP Descriptions](#).

Table 1-6 Accounting-Request START Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
User-Name	1	String	Optional
NAS-IP-Address	4	IPv4	Conditional (NAS-IP-Address or NAS-Identifier must be present)
Service-Type	6	Framed	Optional
Framed-Protocol	7	7 (GPRS PDP Context)	Optional
Framed-IP-Address	8	IPv4	Conditional
Class	25	String	Conditional (Present if received in Access-Request message)
Called-Station-Id	30	APN (UTF-8 encoded characters)	Mandatory
Calling-Station-Id	31	MSISDN in international format (UTF8 encoded decimal)	Optional

Table 1-6 Accounting-Request START Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
NAS-Identifier	32	String	Conditional (NAS-IP-Address or NAS-Identifier must be present)
Acct-Status-Type	40	START	Mandatory
Acct-Delay-Time	41	32 Bit Unsigned Integer	Optional
Acct-Session-Id	44	GGSN/PGW IP Address (IPv4 or IPv6) and Charging-ID concatenated in a UTF-8 encoded hexadecimal character	Mandatory
Acct-Authentic	45	RADIUS or LOCAL	Optional
Acct-Multi-Session-Id	50	String	Optional
Event-Timestamp	55	64 Bit Unsigned Integer	Optional
NAS-Port-Type	61	As Per RFC 2865 [38]	Optional
Tunnel-Type	64	32 Bit Unsigned Integer	Optional
Tunnel-Medium-Type	65	32 Bit Unsigned Integer	Optional
Tunnel-Client-Endpoint	66	String	Optional
Tunnel-Server-Endpoint	67	String	Optional
Acct-Tunnel-Connection	68	String	Optional
Tunnel-Password	69	String	Optional
Tunnel-Assignment-ID	82	String	Optional
Tunnel-Preference	83	32 Bit Unsigned Integer	Optional
NAS-Port-Id	87	String	Optional
Tunnel-Client-Auth-ID	90	String	Optional

Table 1-6 Accounting-Request START Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
Tunnel-Server-Auth-ID	91	String	Optional
NAS-IPv6-Address	95	IPv6	Conditional
Framed-Interface-ID	96	64 Bits As Per IETF RFC 3162	Optional
Framed-IPv6-Prefix	97	IPv6	Conditional
Delegated-IPv6-Prefix	123	IPv6	Conditional
3GPP Vendor-Specific	26/10415	As per section 16.4.7.2 of 3GPP TS 29.061 V11.0.0 (2012-03)	
3GPP-IMSI	1	String	Optional
3GPP-Charging-Id	2	Integer	Optional
3GPP-PDP-Type	3	Integer	Conditional (Mandatory if Framed-Protocol is present)
3GPP-CG-Address	4	IPv4	Optional
3GPP-GPRS-Negotiated-QoS-Profile	5	String	Optional
3GPP-SGSN-Address	6	IPv4	Optional
3GPP-GGSN-Address	7	IPv4	Optional
3GPP-IMSI-MCC-MNC	8	String	Optional
3GPP-GGSN-MCC-MNC	9	String	Optional
3GPP-NSAPI	10	String	Optional
3GPP-Selection-Mode	12	String	Optional
3GPP-Charging-Characteristics	13	String	Optional
3GPP-CG-IPv6-Address	14	IPv6	Optional
3GPP-SGSN-IPv6-Address	15	IPv6	Optional
3GPP-GGSN-IPv6-Address	16	IPv6	Optional
3GPP-IPv6-DNS-Servers	17	IPv6	Optional
3GPP-SGSN-MCC-MNC	18	String	Optional
3GPP-Teardown-Indicator	19	String	Optional

Table 1-6 Accounting-Request START Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
3GPP-IMEISV	20	String	Optional
3GPP-RAT-Type	21	String	Optional
3GPP-User-Location-Info	22	String	Optional
3GPP-MS-TimeZone	23	String	Optional
3GPP-CAMEL-Charging-Info	24	String	Optional
3GPP-Packet-Filter	25	String	Optional
3GPP-Negotiated-DSCP	26	String	Optional

Table 1-7 lists the VSAs for Accounting-Request START messages.

Table 1-7 Vendor-Specific Accounting-Request START Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
Huawei ePDG UE Local IP	26/2011/205	IPv4	Optional
SN-Ecs-Data-Volume	26/8164/176	Unsigned Integer	Optional
Sub-Profile-Name	26/2352/91	String	Optional
TMO-Charging-Rule-Base-Name	26/29168/1	String	Optional
TMO-Virtual-Gi-ID	26/29168/15	32 Bit Unsigned Integer	Optional

Accounting-Request STOP

Table 1-8 lists the Accounting-Request STOP message attributes. For a description of each attribute, see [Chapter 2, RADIUS AVP Descriptions](#).

Table 1-8 Accounting-Request STOP Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
User-Name	1	String	Optional
NAS-IP-Address	4	IPv4	Conditional (NAS-IP-Address or NAS-Identifier must be present)
Service-Type	6	Framed	Optional
Framed-Protocol	7	7 (GPRS PDP Context)	Optional
Framed-IP-Address	8	IPv4	Conditional
Class	25	String	Optional
Called-Station-Id	30	APN (UTF-8 encoded characters)	Mandatory
Calling-Station-Id	31	MSISDN in international format (UTF8 encoded decimal)	Optional
NAS-Identifier	32	String	Conditional (NAS-IP-Address or NAS-Identifier must be present)
Acct-Status-Type	40	STOP	Mandatory
Acct-Delay-Time	41	32 Bit Unsigned Integer	Optional
Acct-Input-Octets	42	32 Bit Unsigned Integer	Optional
Acct-Output-Octets	43	32 Bit Unsigned Integer	Optional

Table 1-8 Accounting-Request STOP Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
Acct-Session-Id	44	GGSN/PGW IP Address (IPv4 or IPv6) and Charging-ID concatenated in a UTF-8 encoded hexadecimal character	Mandatory
Acct-Authentic	45	RADIUS or LOCAL	Optional
Acct-Session-Time	46	Seconds	Optional
Acct-Input-Packets	47	Packets	Optional
Acct-Output-Packets	48	Packets	Optional
Acct-Terminate-Cause	49	As Per RFC 2866 [39]	Optional
Acct-Multi-Session-Id	50	String	Optional
Acct-Input-Gigawords	52	32 Bit Unsigned Integer	Optional
Acct-Output-Gigawords	53	32 Bit Unsigned In	Optional
Event-Timestamp	55	64 Bit Unsigned Integer	Optional
NAS-Port-Type	61	As Per RFC 2865 [38]	Optional
Tunnel-Type	64	32 Bit Unsigned Integer	Optional
Tunnel-Medium-Type	65	32 Bit Unsigned Integer	Optional
Tunnel-Client-Endpoint	66	String	Optional
Tunnel-Server-Endpoint	67	String	Optional
Acct-Tunnel-Connection	68	String	Optional
Tunnel-Password	69	String	Optional
Tunnel-Preference	83	32 Bit Unsigned Integer	Optional
NAS-Port-Id	87	String	Optional

Table 1-8 Accounting-Request STOP Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
Tunnel-Client-Auth-ID	90	String	Optional
Tunnel-Server-Auth-ID	91	String	Optional
NAS-IPv6-Address	95	IPv6	Conditional
Framed-Interface-ID	96	64 Bits As Per IETF RFC 3162 [50]	Optional
Framed-IPv6-Prefix	97	IPv6	Conditional
Delegated-IPv6-Prefix	123	IPv6	Conditional
3GPP Vendor-Specific	26/10415	As per section 16.4.7.2 of 3GPP TS 29.061 V11.0.0 (2012-03)	
3GPP-IMSI	1	String	Optional
3GPP-Charging-Id	2	Integer	Optional
3GPP-PDP-Type	3	Integer	Conditional (Mandatory if Framed-Protocol is present)
3GPP-CG-Address	4	IPv4	Optional
3GPP-GPRS-Negotiated-QoS-Profile	5	String	Optional
3GPP-SGSN-Address	6	IPv4	Optional
3GPP-GGSN-Address	7	IPv4	Optional
3GPP-IMSI-MCC-MNC	8	String	Optional
3GPP-GGSN-MCC-MNC	9	String	Optional
3GPP-NSAPI	10	String	Optional
3GPP-Session-Stop-Indicator	11	String	Optional
3GPP-Selection-Mode	12	String	Optional
3GPP-Charging-Characteristics	13	String	Optional
3GPP-CG-IPv6-Address	14	IPv6	Optional
3GPP-SGSN-IPv6-Address	15	IPv6	Optional
3GPP-GGSN-IPv6-Address	16	IPv6	Optional

Table 1-8 Accounting-Request STOP Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
3GPP-IPv6-DNS-Servers	17	IPv6	Optional
3GPP-SGSN-MCC-MNC	18	String	Optional
3GPP-TearDown-Indicator	19	String	Optional
3GPP-IMEISV	20	String	Optional
3GPP-RAT-Type	21	String	Optional
3GPP-User-Location-Info	22	String	Optional
3GPP-MS-TimeZone	23	String	Optional
3GPP-Packet-Filter	25	String	Optional
3GPP-Negotiated-DSCP	26	String	Optional

[Table 1-9](#) lists the VSAs for Accounting-Request STOP messages.

Table 1-9 Vendor-Specific Accounting-Request STOP Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
Huawei ePDG UE Local IP	26/2011/205	IPv4	Optional
SN-Ecs-Data-Volume	26/8164/176	Unsigned Integer	Optional
TMO-Charging-Rule-Base-Name	26/29168/1	String	Optional
TMO-Virtual-Gi-ID	26/29168/15	32 Bit Unsigned Integer	Optional

Accounting-Request INTERIM

The GGSN/PGW sends the Accounting-Request INTERIM message to the RADIUS server to indicate that the parameters have changed during an active session, such as a RAT change, SGSN/SGW address change, or QoS change.

The supported Accounting-Request INTERIM message attributes are the same as the Accounting-Request STOP attributes, except the Acct-Status-Type attribute contains a value of *Interim-Update* instead of *STOP*.

Accounting-Request ON

The GGSN/PGW sends an Accounting-Request ON message to the RADIUS server to indicate that the MCC device is up after a scheduled reboot and to ensure session information between the GGSN/PGW and the RADIUS server is synchronized.

[Table 1-10](#) lists the Accounting-Request ON message attributes. For a description of each attribute, see [Chapter 2, RADIUS AVP Descriptions](#).

Table 1-10 Accounting-Request ON Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
NAS-IP-Address	4	IPv4	Conditional (NAS-IP-Address or NAS-Identifier must be present)
Called-Station-Id	30	APN (UTF-8 encoded characters)	Optional
NAS-Identifier	32	String	Conditional (NAS-IP-Address or NAS-Identifier must be present)
Acct-Status-Type	40	ON	Mandatory
NAS-IPv6-Address	95	IPv6	Conditional (NAS-IPv6-Address or NAS-Identifier must be present)

Accounting-Request OFF

The GGSN/PGW sends an Accounting-Request OFF message to the RADIUS server to indicate that the MCC device is going down due to a scheduled reboot and to ensure session information between the GGSN/PGW and the RADIUS server is synchronized.

[Table 1-11](#) lists the Accounting-Request OFF message attributes. For a description of each attribute, see [Chapter 2, RADIUS AVP Descriptions](#).

Table 1-11 Accounting-Request OFF Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Requirement
NAS-IP-Address	4	IPv4	Conditional (NAS-IP-Address or NAS-Identifier must be present)
Called-Station-Id	30	APN (UTF-8 encoded characters)	Optional
NAS-Identifier	32	String	Conditional (NAS-IP-Address or NAS-Identifier must be present)
Acct-Status-Type	40	OFF	Mandatory
NAS-IPv6-Address	95	IPv6	Conditional (NAS-IPv6-Address or NAS-Identifier must be present)

Accounting-Response

Upon receiving an Accounting-Request message, the RADIUS server sends an Accounting-Response to the GGSN/PGW to acknowledge the successful receipt of the Accounting-Request (START/INTERIM/STOP/ON/OFF).

Disconnect-Request

The remote RADIUS server sends a Disconnect-Request to the GGSN/PGW to terminate a user session after the user session has been authenticated and authorized to access the requested network services.

Table 1-12 lists the supported Disconnect-Request message attributes according to RFC 5176.

Table 1-12 Disconnect-Request Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Description
NAS Identification Attributes			
NAS-IP-Address	4	IPv4	The IPv4 address of the NAS.
NAS-Identifier	32	String	String identifying the NAS.
NAS-IPv6-Address	95	IPv6	The IPv6 address of the NAS.
Session Identification Attributes			
User-Name	1	String	The name of the user associated with one or more sessions.
NAS-Port	5	Integer	The port on which a session is terminated.
Framed-IP-Address	8	IPv4	The IPv4 address associated with a session.
Proxy-State	33	String	If there are Proxy-State attributes in a Disconnect-Request, the RADIUS client must include those Proxy-State attributes in the Disconnect-ACK or Disconnect-NAK response to the RADIUS server. (Defined in RFC 5176)
Vendor-Specific	26		One or more vendor-specific identification attributes.
Called-Station-Id	30	APN (UTF-8 encoded characters)	The link address to which a session is connected.
Calling-Station-Id	31	MSISDN in international format (UTF8 encoded decimal)	The link address from which one or more sessions are connected.

Table 1-12 Disconnect-Request Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Description
Acct-Session-Id	44	GGSN/PGW IP Address (IPv4 or IPv6) and Charging-ID concatenated in a UTF-8 encoded hexadecimal character	The identifier uniquely identifying a session on the NAS.
NAS-Port-Id	87	String	Identifies the port that is authenticating the user.
Framed-IPv6-Prefix	97	IPv6	The IPv6 prefix associated with a session, always sent with Framed-Interface-Id.

Disconnect-ACK

If the PGW/GGSN successfully terminates the user session(s) and discards all associated session context, it responds to the Disconnect-Request message with a Disconnect-ACK message.

Table 1-13 Disconnect-ACK Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Description
Proxy-State	33	String	If there are Proxy-State attributes in a Disconnect-Request, the RADIUS client must include those Proxy-State attributes in the Disconnect-ACK or Disconnect-NAK response to the RADIUS server.

Disconnect-NAK

If the PGW/GGSN is unable to disconnect one or more sessions and discard all associated session context, it responds to the Disconnect-Request message with a Disconnect-NAK.

Table 1-14 Disconnect-NAK Message Attributes

Attribute Name	Attribute Code	Attribute Data Type	Description
Proxy-State	33	String	If there are Proxy-State attributes in a Disconnect-Request, the RADIUS client must include those Proxy-State attributes in the Disconnect-ACK or Disconnect-NAK response to the RADIUS server.

RADIUS AVP Descriptions

This section describes the following supported Attribute Value Pair (AVP) definitions in alphabetical order for the RADIUS interfaces:

- [RADIUS AVP Definitions \(page 2-2\)](#)
- [3GPP RADIUS AVP Definitions \(page 2-18\)](#)
- [Vendor-Specific RADIUS AVP Definitions \(page 2-27\)](#)

***Note:** Refer to the 3GPP specifications and RFCs listed in [References and Specifications \(page -xviii\)](#) for more details.*

RADIUS AVP Definitions

This section describes the following AVPs used in RADIUS messages.

- Acct-Authentic (page 2-3)
- Acct-Delay-Time (page 2-3)
- Acct-Input-Gigawords (page 2-3)
- Acct-Input-Octets (page 2-3)
- Acct-Input-Packets (page 2-4)
- Acct-Multi-Session-Id (page 2-4)
- Acct-Output-Gigawords (page 2-4)
- Acct-Output-Octets (page 2-4)
- Acct-Output-Packets (page 2-5)
- Acct-Session-Id (page 2-5)
- Acct-Session-Time (page 2-5)
- Acct-Status-Type (page 2-6)
- Acct-Terminate-Cause (page 2-6)
- Acct-Tunnel-Connection (page 2-6)
- Called-Station-Id (page 2-6)
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- Class (page 2-8)
- CHAP-Challenge (page 2-7)
- CHAP-Password (page 2-7)
- Chargeable-Unit-Identity (page 2-8)
- Class (page 2-8)
- Delegated-IPv6-Prefix (page 2-8)
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- Framed-IPv6-Prefix (page 2-10)
- Framed-IPv6-Route (page 2-10)
- Framed-MTU (page 2-10)
- Framed-Pool (page 2-10)
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- Idle-Timeout (page 2-11)
- NAS-Identifier (page 2-11)
- NAS-IP-Address (page 2-12)
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- Tunnel-Assignment-ID (page 2-14)
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- Tunnel-Password (page 2-15)
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- Tunnel-Server-Auth-ID (page 2-15)
- Tunnel-Server-Endpoint (page 2-16)
- Tunnel-Type (page 2-16)
- User-Name (page 2-16)
- User-Password (page 2-17)

Acct-Authentic

AVP Name	Acct-Authentic
AVP Code	45
AVP Data Type	RADIUS or LOCAL
Description	Indicates how the user was authenticated.

Acct-Delay-Time

AVP Name	Acct-Delay-Time
AVP Code	41
AVP Data Type	32 Bit Unsigned Integer
Description	Indicates how many seconds the GGSN/PGW has been trying to send a particular record. The value of this attribute can be subtracted from the time of arrival on the server to determine the approximate time of the event generating this Accounting-Request.

Acct-Input-Gigawords

AVP Name	Acct-Input-Gigawords
AVP Code	52
AVP Data Type	32 Bit Unsigned Integer
Description	Indicates the number of times that the Acct-Input-Octets counter has wrapped around 2^{32} .

Acct-Input-Octets

AVP Name	Acct-Input-Octets
AVP Code	42
AVP Data Type	32 Bit Unsigned Integer
Description	Indicates how many octets have been sent by the user for the IP-CAN bearer.

Acct-Input-Packets

AVP Name	Acct-Input-Packets
AVP Code	47
AVP Data Type	Packet
Description	Indicates how many packets have been sent by the user.

Acct-Multi-Session-Id

AVP Name	Acct-Multi-Session-Id
AVP Code	50
AVP Data Type	String
Description	Indicates a unique Accounting ID to link together multiple related sessions in a log file. Each session linked together has a unique Acct-Session-Id but the same Acct-Multi-Session-Id. The Acct-Multi-Session-Id attribute contains the Acct-Session-Id of the default bearer.

Acct-Output-Gigawords

AVP Name	Acct-Output-Gigawords
AVP Code	53
AVP Data Type	32 Bit Unsigned Integer
Description	Indicates the number of times that the Acct-Output-Octets counter has wrapped around 2^{32} .

Acct-Output-Octets

AVP Name	Acct-output-Octets
AVP Code	43
AVP Data Type	32 Bit Unsigned Integer
Description	Indicates how many octets have been received by the user for the IP-CAN bearer.

Acct-Output-Packets

AVP Name	Acct-Output-Packets
AVP Code	48
AVP Data Type	Packet
Description	Indicates how many packets have been received by the user.

Acct-Session-Id

AVP Name	Acct-Session-Id
AVP Code	44
AVP Data Type	String
Description	Contains a unique user session identifier. The GGSN/PGW IP address is the same as the IP address used in the CDRs generated by the GGSN/PGW. To exclude/include the PDN session start time in the RADIUS Acct-Session-Id sent in RADIUS Access-Request and Accounting-Request messages, see Configuring the RADIUS Authorization Service (page 1-15) and Configuring the RADIUS Accounting Service (page 1-18).

Acct-Session-Time

AVP Name	Acct-Session-Time
AVP Code	46
AVP Data Type	Seconds
Description	Indicates the duration of the user session in seconds. Only present in Accounting-Request Stop records.

Acct-Status-Type

AVP Name	Acct-Status-Type
AVP Code	40
AVP Data Type	START/STOP/INTERIM/ON/OFF
Description	Indicates the Accounting-Request status: ■ START – Begins a user session. ■ INTERIM – Updates a user session. ■ STOP – Ends a user session. ■ ON – Indicates that the MCC device is up. ■ OFF – Indicates that the MCC device is down.

Acct-Terminate-Cause

AVP Name	Acct-Terminate-Cause
AVP Code	49
AVP Data Type	As per RFC 2866
Description	Indicates how the session was terminated.

Acct-Tunnel-Connection

AVP Name	Acct-Tunnel-Connection
AVP Code	68
AVP Data Type	String
Description	Indicates the identifier assigned to the tunnel session.

Called-Station-Id

AVP Name	Called-Station-Id
AVP Code	30
AVP Data Type	Access Point Name (APN) (UTF-8 Encoded Characters)
Description	Identifies the network that the subscriber is attempting to use.

Calling-Station-Id

AVP Name	Calling-Station-Id
AVP Code	31
AVP Data Type	MSISDN in international format according to 3GPP TS 23.003 [40], UTF-8 encoded decimal character
Description	Identifies the Mobile Station (MS) from which the session was initiated. This attribute is configurable on a per APN basis. <i>Note: There are no leading characters in front of the country code.</i>

CHAP-Challenge

AVP Name	CHAP-Challenge
AVP Code	60
AVP Data Type	String
Description	The user provides the Challenge Handshake Authentication Protocol (CHAP) Challenge (extracted from the Protocol Configuration Options field of the Create PDP Context Request message) in the RADIUS Access-Request message if the authentication protocol is CHAP. If the PDP context uses a Point-to-Point Protocol (PPP) connection, the CHAP Challenge is provided during the PPP authentication phase.

CHAP-Password

AVP Name	CHAP-Password
AVP Code	3
AVP Data Type	String
Description	The user provides the CHAP password (extracted from the Protocol Configuration Options field of the Create PDP Context Request message) in the RADIUS Access-Request message if the authentication protocol is CHAP. If the PDP context uses a Point-to-Point Protocol (PPP) connection, the CHAP password is provided during the PPP authentication phase.

Chargeable-Unit-Identity

AVP Name	Chargeable-Unit-Identity
AVP Code	89
AVP Data Type	String
Description	When Option 82 information (circuit-id/remote-id/vendor-id) is present in the DHCP Discover/DHCP Request message, the MCC gateway sends this information via the RADIUS Access-Request message in the Chargeable-User-Identity attribute, as defined in RFC 4372.

Class

AVP Name	Class
AVP Code	25
AVP Data Type	String
Description	Indicates the identifier used in all subsequent accounting messages.

Delegated-IPv6-Prefix

AVP Name	Delegated-IPv6-Prefix
AVP Code	123
AVP Data Type	IPv6
Description	Indicates the IPv6 prefix delegated to the user.

Event-Timestamp

AVP Name	Event-Timestamp
AVP Code	55
AVP Data Type	64 Bit Unsigned Integer
Description	Records the time that this event occurred on the NAS, in seconds since January 1, 1970 00:00 UTC.

Framed-Interface-ID

AVP Name	Framed-Interface-Id
AVP Code	96
AVP Data Type	64 Bits As Per IETF RFC 3162 [50]
Description	Indicates the IPv6 Interface Identifier provided by the GGSN/PGW to the UE at Initial Attach.

Framed-IP-Address

AVP Name	Framed-IP-Address
AVP Code	8
AVP Data Type	IPv4
Description	Indicates the allocated IPv4 address for the user if the RADIUS server is used to allocate IP addresses.

Framed-IP-Netmask

AVP Name	Framed-IP-Netmask
AVP Code	9
AVP Data Type	IPv4
Description	Indicates the allocated IP netmask for the user if the RADIUS server is used to allocate the IP netmask.

Framed-IPv6-Pool

AVP Name	Framed-IPv6-Pool
AVP Code	100
AVP Data Type	String
Description	Indicates the name of the IPv6 prefix pool for the specific APN.

Framed-IPv6-Prefix

AVP Name	Framed-IPv6-Prefix
AVP Code	97
AVP Data Type	IPv6
Description	Indicates the allocated IPv6 address for the user if the RADIUS server is used to allocate IP addresses.

Framed-IPv6-Route

AVP Name	Framed-IPv6-Route
AVP Code	99
AVP Data Type	String
Description	Provides routing information to be configured for the user on the network access server (NAS) as defined in Internet Engineering Task Force (IETF) standard RFC 3162. This attribute can appear multiple times (MCC supports a maximum of 16) in Access-Accept packets.

Framed-MTU

AVP Name	Framed-MTU
AVP Code	12
AVP Data Type	String
Description	Indicates the Maximum Transmission Unit (MTU) of the packet data protocol protocol data units (PDP PDUs), between the mobile station (MS) and GGSN/PGW.

Framed-Pool

AVP Name	Framed-Pool
AVP Code	88
AVP Data Type	String
Description	Indicates the name of the configured local ip-sub-pool name used to allocate the UE IP address. <i>Note: Applies if the <code>ue-ip-allocation-mode</code> parameter on the Workflow Control Profile is set to <code>radius-ue-ip-allocation</code>.</i>

Framed-Protocol

AVP Name	Framed-Protocol
AVP Code	7
AVP Data Type	7 (GPRS PDP Context)
Description	Indicates the framing protocol to be used for the user.

Framed-Route

AVP Name	Framed-Route
AVP Code	22
AVP Data Type	String
Description	Provides routing information to be configured for the user on this network access server (NAS) as defined in Internet Engineering Task Force (IETF) standard RFC 2865. This attribute can appear multiple times (MCC supports a maximum of 16) in Access-Accept packets. The MCC ignores the Framed-Route AVP with missing metrics.

Idle-Timeout

AVP Name	Idle-Timeout
AVP Code	28
AVP Data Type	32 Bit Unsigned Integer
Description	Indicates the amount of idle time (in seconds) after which the user session terminates.

NAS-Identifier

AVP Name	NAS-Identifier
AVP Code	32
AVP Data Type	String
Description	The hostname of the GGSN/PGW requesting user authentication.

NAS-IP-Address

AVP Name	NAS-IP-Address
AVP Code	4
AVP Data Type	IPv4
Description	The IP address of the GGSN/PGW requesting user authentication.

NAS-IPv6-Address

AVP Name	NAS-IPv6-Address
AVP Code	95
AVP Data Type	IPv6
Description	The IPv6 address of the GGSN/PGW requesting user authentication.

NAS-Port

AVP Name	NAS-Port
AVP Code	5
AVP Data Type	Integer
Description	The port on which a session is terminated.

NAS-Port-Id

AVP Name	NAS-Port-Id
AVP Code	5
AVP Data Type	String
Description	Identifies the NAS port that is authenticating the user.

NAS-Port-Type

AVP Name	NAS-Port-Type
AVP Code	61
AVP Data Type	As Per RFC 2865
Description	Indicates the type of physical port the network access server is using to authenticate the user.

Proxy-State

AVP Name	Proxy-State
AVP Code	33
AVP Data Type	String
Description	If there are any Proxy-State attributes in a Disconnect-Request received from the RADIUS server, the RADIUS client must include those Proxy-State attributes in its response to the RADIUS server. (Defined in RFC 5176)

Service-Type

AVP Name	Service-Type
AVP Code	6
AVP Data Type	2 (Framed) or 17 (Authorize Only)
Description	Indicates the type of service requested or the type of service provided. The Service-Type data type is as follows: ■ <i>Framed</i> – Indicates that a framed protocol, such as PPP or SLIP, should be started for the user. ■ <i>Authorize Only</i> – Applicable only when deferring IPv4 address allocation upon initial connection on a dual-stack IPv4v6 PDN/PDP. In this case, the initial Access-Request message has a Service-Type value of <i>Framed</i>, but the subsequent Access-Request message has a Service-Type value of <i>Authorize Only</i>. Both Access-Request messages must have the <i>3GPP-Allocate-IP-Type</i> sub-attribute.

Session-Timeout

AVP Name	Session-Timeout
AVP Code	27
AVP Data Type	32 Bit Unsigned Integer
Description	Indicates the timeout value (in seconds) after which the user session terminates.

Tunnel-Assignment-ID

AVP Name	Tunnel-Assignment-ID
AVP Code	82
AVP Data Type	String
Description	Used to indicate to the tunnel initiator the particular tunnel to which a session is to be assigned. See Establishing L2TP Tunnel via RADIUS Access-Accept (page 1-33) .

Tunnel-Client-Auth-ID

AVP Name	Tunnel-Client-Auth-ID
AVP Code	90
AVP Data Type	String
Description	Indicates the name used by the tunnel initiator during the authentication phase of tunnel establishment.

Tunnel-Client-Endpoint

AVP Name	Tunnel-Client-Endpoint
AVP Code	66
AVP Data Type	String
Description	Contains the address of the initiator end of the tunnel.

Tunnel-Medium-Type

AVP Name	Tunnel-Medium-Type
AVP Code	65
AVP Data Type	32 Bit Unsigned Integer
Description	Indicates the transport medium used to create a tunnel for protocols that operate over multiple transports, such as L2TP.

Tunnel-Password

AVP Name	Tunnel-Password
AVP Code	69
AVP Data Type	String
Description	Indicates the password used to authenticate to a remote server.

Tunnel-Preference

AVP Name	Tunnel-Preference
AVP Code	83
AVP Data Type	32 Bit Unsigned Integer
Description	If more than one set of tunneling attributes is returned by the RADIUS server to the tunnel initiator, this attribute should be included in each set to indicate the relative preference assigned to each tunnel.

Tunnel-Server-Auth-ID

AVP Name	Tunnel-Server-Auth-ID
AVP Code	91
AVP Data Type	String
Description	Indicates the name used by the tunnel terminator during the authentication phase of tunnel establishment.

Tunnel-Server-Endpoint

AVP Name	Tunnel-Server-Endpoint
AVP Code	67
AVP Data Type	String
Description	Contains the address of the server end of the tunnel.

Tunnel-Type

AVP Name	Tunnel-Type
AVP Code	64
AVP Data Type	32 Bit Unsigned Integer
Description	Indicates the tunneling protocol(s) used.

User-Name

AVP Name	User-Name
AVP Value	1
AVP Data Type	String
Description	The user provides the username (extracted from the Protocol Configuration Options field of the Create PDP Context Request message) in the RADIUS Access-Request message. If the PDP context uses a Point-to-Point Protocol (PPP) connection, the username is provided during the PPP authentication phase. If no username is provided, a generic username, configurable on a per APN basis, is used. <i>Note: You can configure the GGSN/PGW to suppress trailing NULL characters in Protocol Configuration Options (PCO) PAP username/password fields before sending to the AAA server in the RADIUS Access-Request message. For more information, see Suppressing NULL Characters in PCO PAP Username/Password Fields (page 1-50).</i>

User-Password

AVP Name	User-Password
AVP Code	2
AVP Data Type	String
Description	The user provides the user password (extracted from the Protocol Configuration Options field of the Create PDP Context Request message) in the RADIUS Access-Request message if the authentication protocol is Password Authentication Protocol (PAP). If the PDP context uses a Point-to-Point Protocol (PPP) connection, the user password is provided during the PPP authentication phase. If no password is provided, a generic password, configurable on a per APN basis, is used. Note: You can configure the GGSN/PGW to suppress trailing NULL characters in Protocol Configuration Options (PCO) PAP username/password fields before sending to the AAA server in the RADIUS Access-Request message. For more information, see Suppressing NULL Characters in PCO PAP Username/Password Fields (page 1-50).

3GPP RADIUS AVP Definitions

This section describes the supported 3GPP AVPs used in RADIUS messages. The AVPs include Vendor-ID of 26/10415 (3GPP).

- 3GPP-Allocate-IP-Type (page 2-19)
- 3GPP-CAMEL-Charging-Info (page 2-19)
- 3GPP-CG-Address (page 2-19)
- 3GPP-CG-IPv6-Address (page 2-19)
- 3GPP-Charging-Characteristics (page 2-20)
- 3GPP-Charging-Id (page 2-20)
- 3GPP-GGSN-Address (page 2-20)
- 3GPP-GGSN-IPv6-Address (page 2-21)
- 3GPP-GGSN-MCC-MNC (page 2-21)
- 3GPP-GPRS-Negotiated-QoS-Profile (page 2-21)
- 3GPP-IMEISV (page 2-22)
- 3GPP-IMSI (page 2-22)
- 3GPP-IMSI-MCC-MNC (page 2-22)
- 3GPP-IPv6-DNS-Servers (page 2-22)
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- 3GPP-NSAPI (page 2-23)
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- 3GPP-PDP-Type (page 2-24)
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- 3GPP-Session-Stop-Indicator (page 2-24)
- 3GPP-SGSN-Address (page 2-25)
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- 3GPP-Tear-down-Indicator (page 2-26)

3GPP-Allocate-IP-Type

AVP Name	3GPP-Allocate-IP-Type
AVP Code	27
AVP Data Type	String
Description	Indicates whether the Access-Request message is sent for user authentication only and/or for allocation of an IPv4 address and/or for allocation of an IPv6 prefix.

3GPP-CAMEL-Charging-Info

AVP Name	3GPP-CAMEL-Charging-Info
AVP Code	24
AVP Data Type	String
Description	Contains Customized Application of Mobile Enhanced Logic (CAMEL) charging information.

3GPP-CG-Address

AVP Name	3GPP-CG-Address
AVP Code	4
AVP Data Type	IPv4
Description	Indicates the IP address for the charging gateway.

3GPP-CG-IPv6-Address

AVP Name	3GPP-CG-IPv6--Address
AVP Code	14
AVP Data Type	IPv6
Description	Indicates the IPv6 address for the charging gateway.

3GPP-Charging-Characteristics

AVP Name	3GPP-Charging-Characteristics
AVP Code	13
AVP Data Type	String
Description	For the GGSN, this attribute contains the charging characteristics for this PDP context received in the Create PDP Context Request message (only available in R99 and later releases). For the PGW, this attribute contains the charging characteristics for the IP-CAN bearer.

3GPP-Charging-Id

AVP Name	3GPP-Charging-Id
AVP Code	2
AVP Data Type	Integer
Description	For the GGSN, this attribute indicates the charging ID for this PDP Context. The Charging ID together with the GGSN IP address constitutes a unique identifier for the PDP context. For the PGW, this attribute indicates the charging ID for this IP-CAN bearer. The Charging ID together with the PGW IP address constitutes a unique identifier for the IP-CAN bearer.

3GPP-GGSN-Address

AVP Name	3GPP-GGSN-Address
AVP Code	7
AVP Data Type	IPv4
Description	For the GGSN, this attribute represents the GGSN IPv4 address used by the GTP control plane to establish the PDP context. For the PGW, this attribute represents the PGW IPv4 address used on S5/S8, S2a or S2c control plane to establish the IP-CAN session. In both cases, this address is the same as the GGSN/PGW IPv4 address used in CDRs generated by the GGSN/PGW.

3GPP-GGSN-IPv6-Address

AVP Name	3GPP-GGSN-IPv6-Address
AVP Code	16
AVP Data Type	IPv6
Description	For the GGSN, this attribute represents the GGSN IPv6 address that is used by the GTP control plane to establish the PDP context. For the PGW, this attribute represents the PGW IPv6 address used on S5/S8, S2a or S2c control plane to establish the IP-CAN session.

3GPP-GGSN-MCC-MNC

AVP Name	3GPP-GGSN-MCC-MNC
AVP Code	9
AVP Data Type	String
Description	The Mobile Country Code (MCC) and Mobile Network Code (MNC) of the network to which the GGSN/PGW belongs. When the PLMN ID (MCC + MNC), received in the subscriber's IMSI, matches a PLMN ID configured in the home-plmnid-list, the configured PLMN ID is sent in the 3GPP-GGSN-MCC-MNC AVP on all external interfaces (RADIUS, Gx, Gy, Gz/Rf). If an entry is not found in the list, the first entry in the home-plmnid-list is used in the 3GPP-GGSN-MCC-MNC AVP.

3GPP-GPRS-Negotiated-QoS-Profile

AVP Name	3GPP-GPRS-Negotiated-QoS-Profile
AVP Code	5
AVP Data Type	String
Description	For the GGSN, this attribute represents the QoS profile for the PDP context. For the PGW, this attribute represents the QoS profile for the EPS bearer and the authorized APN-AMBR.

3GPP-IMEISV

AVP Name	3GPP-IMEISV
AVP Code	20
AVP Data Type	String
Description	The user International Mobile Equipment Identity of the equipment served (IMEISV) and its software version.

3GPP-IMSI

AVP Name	3GPP-IMSI
AVP Code	1
AVP Data Type	String
Description	The International Mobile Subscriber Identity (IMSI) for this user.

3GPP-IMSI-MCC-MNC

AVP Name	3GPP-IMSI-MCC-MNC
AVP Code	8
AVP Data Type	String
Description	Indicates the Mobile Country Code (MCC) and Mobile Network Code (MNC) extracted from the user's IMSI (first 5 or 6 digits, as applicable from the presented IMSI).

3GPP-IPv6-DNS-Servers

AVP Name	3GPP-IPv6-DNS-Servers
AVP Code	17
AVP Data Type	IPv6
Description	Lists the IPv6 addresses of DNS servers for an APN.

3GPP-MS-TimeZone

AVP Name	3GPP-MS-TimeZone
AVP Code	23
AVP Data Type	String
Description	The Time Zone field and the Daylight Saving Time fields are used to indicate the offset between universal time and local time in steps of 15 minutes of where the MS/UE currently resides.

3GPP-Negotiated-DSCP

AVP Name	3GPP-Negotiated-DSCP
AVP Code	26
AVP Data Type	String
Description	Indicates the DSCP used to mark the IP packets of this PDP context on the Gi interface or EPS Bearer context on the SGi interface.

3GPP-NSAPI

AVP Name	3GPP-NSAPI
AVP Code	10
AVP Data Type	String
Description	For the GGSN, this attribute identifies a particular PDP context for the associated PDN and Mobile Subscriber Integrated Services Digital Network-Number (MSISDN)/IMSI from creation to deletion. For the PGW, this attribute identifies the EPS Bearer ID if is known to the PGW. For example, if the GTP-based S5/S8 is in use.

3GPP-Packet-Filter

AVP Name	3GPP-Packet-Filter
AVP Code	25
AVP Data Type	String
Description	Identifies the packet filter information used for the PDP context or EPS bearer.

3GPP-PDP-Type

AVP Name	3GPP-PDP-Type
AVP Code	3
AVP Data Type	Integer
Description	For the GGSN, this attribute indicates the type of PDP context. For example, IP or PPP. For the PGW, this attribute indicates the PDN Type, for example, IPv4, IPv6, or IPv4v6.

3GPP-RAT-Type

AVP Name	3GPP-RAT-Type
AVP Code	21
AVP Data Type	String
Description	The Radio Access Technology currently serving the User Equipment (UE).

3GPP-Selection-Mode

AVP Name	3GPP-Selection-Mode
AVP Code	12
AVP Data Type	String
Description	For the GGSN, this attribute indicates the selection mode for this PDP context received in the create PDP context request message. For the PGW, this attribute indicates the selection mode for this EPS Bearer received in the Create Session Request Message.

3GPP-Session-Stop-Indicator

AVP Name	3GPP-Session-Stop-Indicator
AVP Code	11
AVP Data Type	String
Description	Indicates to the AAA server that the last PDP context or EPS Bearer of a session is released and the IP-CAN session has been terminated.

3GPP-SGSN-Address

AVP Name	3GPP-SGSN-Address
AVP Code	6
AVP Data Type	IPv4
Description	For the GGSN, this attribute represents the Serving GPRS Support Node (SGSN) IPv4 address used by the GTP control plane to handle control messages. For the PGW, this attribute represents the SGW IPv4 address, trusted non-3GPP IP access, or ePDG used on S5/S8, S2a or S2b for handling control messages.

3GPP-SGSN-IPv6-Address

AVP Name	3GPP-SGSN-IPv6-Address
AVP Code	15
AVP Data Type	IPv6
Description	For the GGSN, this attribute represents the SGSN IPv6 address used by the GPRS Tunneling Protocol (GTP) control plane for handling control messages. For the PGW, this attribute represents the IPv6 address of the SGW, trusted non-3GPP IP access or ePDG used on S5/S8, S2a or S2b for the handling of control messages. It may be used to identify the PLMN to which the user is attached.

3GPP-SGSN-MCC-MNC

AVP Name	3GPP-SGSN-MCC-MNC
AVP Code	18
AVP Data Type	String
Description	For a GGSN/PGW connected to a Gn/Gp SGSN, this attribute represents the MCC and MNC extracted from the RAI within the Create PDP Context Request or Update PDP Context Request message. For a PGW (in GTP/PMIP S5/S8), this attribute represents the MCC and MNC extracted from the Serving Network.

3GPP-User-Location-Info

AVP Name	3GPP-User-Location-Info
AVP Code	22
AVP Data Type	String
Description	Indicates details of where the user equipment is currently located (for example, SAI or CGI).

3GPP-Teardown-Indicator

AVP Name	3GPP-Teardown-Indicator
AVP Code	19
AVP Data Type	String
Description	Informs the GGSN/PGW that all PDP contexts/IP-CAN bearers for this particular user need to be deleted.

Vendor-Specific RADIUS AVP Definitions

This section describes the supported vendor-specific attributes (VSAs) in RADIUS Access-Accept messages.

- [AF-Advice-of-Charge \(page 2-28\)](#)
- [AF-BwFairApp-Name \(page 2-28\)](#)
- [AF-Charging-Instance-Name \(page 2-28\)](#)
- [AF-Default-QoS-Profile-Name \(page 2-29\)](#)
- [AF-Dhcp-Option82-Circuit-Id \(page 2-29\)](#)
- [AF-Dhcp-Option82-Id \(page 2-29\)](#)
- [AF-Dhcp-Option82-Remote-Id \(page 2-29\)](#)
- [AF-EDR-Interface-Name \(page 2-30\)](#)
- [AF-IP-POOL-Name \(page 2-30\)](#)
- [AF-Msisdn \(page 2-31\)](#)
- [AF-Rule-Base-Name \(page 2-31\)](#)
- [AF-Rule-Name \(page 2-31\)](#)
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- [MS-Primary-DNS-Server \(page 2-34\)](#)
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- [Nokia-OCS-ID1 \(page 2-35\)](#)
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- [SN-Ecs-Data-Volume \(page 2-37\)](#)
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- [3GPP2-Carrier-ID \(page 2-38\)](#)
- [3GPP2-GMT-Time-Zone-Offset \(page 2-39\)](#)

AF-Advice-of-Charge

AVP Name	AF-Advice-Of-Charge
AVP Code	26/37963/22
AVP Data Type	String
Description	Used to place users in either a wifi-open state or wifi-closed state, based on the received values. The following values are defined: ■ 0 (disabled) – Users are allowed full access to the network (wifi-open). ■ 1 (enabled) – All traffic is dropped except for HTTP and DNS (wifi-close). Note: Do not use the AF-Advice-Of-Charge VSA and the AF-SLA-Profile-Id VSA in the same RADIUS message. If both VSAs are used in the same message, the AF-Advice-Of-Charge VSA takes precedence.

AF-BwFairApp-Name

AVP Name	AF-BwFairApp-Name
AVP Code	26/37963/4
AVP Data Type	String
Description	Contains the Affirmed vendor-specific Bandwidth Fair Usage Application. Note: The code field is two octets.

AF-Charging-Instance-Name

AVP Name	AF-Charging-Instance-Name
AVP Code	26/37963/7
AVP Data Type	String
Description	Contains the Affirmed vendor-specific charging instance name ID. Note: The code field is two octets.

AF-Default-QoS-Profile-Name

AVP Name	AF-Default-QoS-Profile-Name
AVP Code	26/37963/6
AVP Data Type	String
Description	Used to update a subscriber's QoS profile. The RADIUS server can send the CoA-Request message multiple times with different QoS profiles. The profile names received in AF-Default-QoS-Profile-Name are pre-configured on the TWAG. Initially, the subscriber is assigned a pre-configured default QoS profile. <i>Note: The code field is two octets.</i>

AF-Dhcp-Option82-Circuit-Id

AVP Name	AF-Dhcp-Option82-Circuit-Id
AVP Code	26/37963/27
AVP Data Type	String
Description	Contains the parsed circuit-id from the Options 82 field in the DHCP message.

AF-Dhcp-Option82-Id

AVP Name	AF-Dhcp-Option82-Id
AVP Code	26/37963/25
AVP Data Type	String
Description	When Option 82 information (circuit-id/remote-id/vendor-id) is present in the DHCP Discover/ DHCP Request message, the MCC gateway sends this information via the RADIUS Access-Request message in the Chargeable-User-Identity attribute, as defined in RFC 4372, and the AF-Dhcp-Option82-Id VSA.

AF-Dhcp-Option82-Remote-Id

AVP Name	AF-Dhcp-Option82-Remote-Id
AVP Code	26/37963/28
AVP Data Type	String
Description	Contains the parsed remote-id from the Options 82 field in the DHCP message.

AF-EDR-Interface-Name

AVP Name	AF-EDR-Interface-Name
AVP Code	26/37963/29
AVP Data Type	String
Description	Contains the Event Data Record (EDR) interface name used to override the EDR interface configured in the Workflow Control profile. When the GGSN/PGW receives the AF-EDR-Interface-Name VSA in the RADIUS Access-Accept message, it replaces the EDR interface configured in the Workflow Control profile with the EDR interface received in the AF-EDR-Interface-Name VSA. The VSA is also used if the EDR interface is not configured under the Control profile. The EDR interface received in the AF-EDR-Interface-Name VSA must match a pre-configured EDR interface on the MCC. If the received EDR interface name does not match an EDR interface configured on the MCC, the configured EDR interface from the Workflow Control profile will be used. The EDR interface received in the AF-EDR-Interface-Name VSA takes precedence over any configured value for the EDR interface. For example, if the Control profile or Subscriber Analyzer is switched, the EDR interface received in the Access-Accept message will still take precedence over the EDR interface configured in the new Control profile. Use the following CLI command to enable the AF-EDR-Interface-Name VSA to be sent in RADIUS Access-Accept messages. The default <code>presentFlag</code> setting is <code>no</code>. <pre>an3000-mcm-slot7cpul(config)# service-construct data-record-template [pgw ggsn] auth radius access-accept <template-name> aff_edrinterfacename presentFlag yes</pre>

AF-IP-POOL-Name

AVP Name	AF-IP-Pool-Name
AVP Code	26/37963/3
AVP Data Type	String
Description	Contains the Affirmed vendor-specific IP address pool name ID. <i>Note: The code field is two octets.</i>

AF-Msisdn

AVP Name	AF-Msisdn
AVP Code	26/37963/26
AVP Data Type	String
Description	Contains an MSISDN value used to set the MSISDN for the specific UE.

AF-Rule-Base-Name

AVP Name	AF-Rule-Base-Name
AVP Code	26/37963/1
AVP Data Type	String
Description	Contains the Affirmed vendor-specific rule base name ID. <i>Note: The code field is two octets.</i>

AF-Rule-Name

AVP Name	AF-Rule-Name
AVP Code	26/37963/2
AVP Data Type	String
Description	Contains the Affirmed vendor-specific rule name ID. <i>Note: The code field is two octets.</i>

AF-SLA-Profile-ID

AVP Name	AF-SLA-Profile-ID
AVP Code	26/37963/21
AVP Data Type	String
Description	Used in CoA-Request messages to place users in either a wifi-open state or wifi-closed state, based on the received values. The RADIUS AF-SLA-Profile-ID VSA contains two values: ■ <code>wifi-open</code> – Users are allowed full access to the network. ■ <code>wifi-close</code> – All traffic is dropped except for HTTP and DNS. Note: Do not use the AF-Advice-Of-Charge VSA and the AF-SLA-Profile-Id VSA in the same RADIUS message. If both VSAs are used in the same message, the AF-Advice-Of-Charge VSA takes precedence.

AF-Workflow-Control-Profile-Name

AVP Name	AF-Workflow-Control-Profile-Name
AVP Code	26/37963/23
AVP Data Type	String
Description	Identifies a pre-configured Workflow Control Profile to be applied to the subscriber when the UE is reattaching to the MCC TWAG. Note: Both the AF-Workflow-Control-Profile-Name and AF-Workflow-Data-Profile-Name VSAs need to be present in the RADIUS Access-Accept message to bypass the Subscriber Analyzer; otherwise the Subscriber Analyzer is used to select the Workflow Control and Data Profiles.

AF-Workflow-Data-Profile-Name

AVP Name	AF-Workflow-Data-Profile-Name
AVP Code	26/37963/24
AVP Data Type	String
Description	Identifies a pre-configured Workflow Data Profile to be applied to the subscriber when the UE is reattaching to the MCC TWAG. Note: Both the AF-Workflow-Control-Profile-Name and AF-Workflow-Data-Profile-Name VSAs need to be present in the RADIUS Access-Accept message to bypass the Subscriber Analyzer; otherwise the Subscriber Analyzer is used to select the Workflow Control and Data Profiles.

Cisco-AV-Pair

AVP Name	Cisco-AV-Pair
AVP Code	26/9/1
AVP Data Type	String
Description	Identifies Cisco vendor-specific attribute-value (AV) Pairs.

Huawei ePDG UE Local IP

AVP Name	Huawei ePDG UE Local IP
AVP Code	26/2011/205
AVP Data Type	IPv4
Description	When the PGW receives the IKEv2 Tunnel IP Address IE (UE local IP address) from the ePDG via the S2b interface, the PGW includes this information within the Huawei ePDG UE Local IP VSA and sends this VSA in all subsequent RADIUS Accounting-Request messages (Start/Interim/Stop) for all bearers belonging to that PDN session. This VSA will be included in Accounting-Request messages for both the primary and mirrored Accounting interfaces. The PGW can receive the IKEv2 Tunnel IP Address IE in any of the following messages: ■ Create Session Request ■ Create Bearer Response ■ Modify Bearer Request ■ Delete Session Request ■ Delete Bearer Response ■ Update Bearer Response The Huawei ePDG UE Local IP VSA is configurable in RADIUS Accounting-Request data record templates. The default <code>presentFlag</code> setting is <code>no</code>, indicating the VSA is not included in RADIUS Accounting-Request messages. Note: Although the IKEv2 Tunnel IP Address IE may carry either an IPv4 or IPv6 address, the PGW will only send the IPv4 address in RADIUS Accounting-Requests and will ignore the IPv6 address.

MS-Primary-DNS-Server

AVP Name	MS-Primary-DNS-Server
AVP Code	26/311/28
AVP Data Type	IPv4
Description	Contains the primary DNS server address for this APN.

MS-Primary-NBNS-Server

AVP Name	MS-Primary-NBNS-Server
AVP Code	26/311/30
AVP Data Type	IPv4
Description	Contains the primary NetBIOS server address for this APN.

MS-Secondary-DNS-Server

AVP Name	MS-Secondary-DNS-Server
AVP Code	26/311/29
AVP Data Type	IPv4
Description	Contains the secondary DNS server address for this APN.

MS-Secondary-NBNS-Server

AVP Name	MS-Secondary-NBNS-Server
AVP Code	26/311/31
AVP Data Type	IPv4
Description	Contains the secondary NetBIOS server address for this APN.

Nokia-OCS-ID1

AVP Name	Nokia-OCS-ID1
AVP Code	26/94/12
AVP Data Type	Integer
Description	Contains the Nokia vendor-specific Online Charging System (OCS) ID1.

Nokia-OCS-ID2

AVP Name	Nokia-OCS-ID2
AVP Code	26/94/13
AVP Data Type	Integer
Description	Contains the Nokia vendor-specific Online Charging System (OCS) ID2.

Nokia-Requested-APN

AVP Name	Nokia-Requested-APN
AVP Code	26/94/15
AVP Data Type	String
Description	Contains the Nokia vendor-specific requested Access Point Name (APN).

Nokia-Session-Charging-Type

AVP Name	Nokia-Session-Charging-Type
AVP Code	26/94/11
AVP Data Type	String
Description	Contains the Nokia vendor-specific session charging type.

TMO-Charging-Rule-Base-Name

AVP Name	TMO-Charging-Rule-Base-Name
AVP Code	26/29168/1
AVP Data Type	String
Description	Supported on the GiGW in incoming RADIUS Accounting messages When this VSA is received in incoming RADIUS Accounting-Request messages, it can be used to select a data profile for GiGW sessions in Subscriber Aware Mode. In Subscriber Aware Mode, subscriber session information is learned from RADIUS Accounting messages via the GiGW session learner (configured with either RADIUS Server or RADIUS Proxy mode). When either the Sub-Profile-Name or the TMO-Charging-Rule-Base-Name VSA is received in an incoming RADIUS Accounting-Request message, the MCC uses the value received in the VSA to select a data profile for the GiGW session and the received value overrides the subscriber analyzer configuration. If an Accounting-Request INTERIM message is received with a data profile name that is different from the one received in the Accounting-Request START message, the MCC switches to the new data profile for that session. <i>Note: The Sub-Profile-Name and TMO-Charging-Rule-Base-Name VSA cannot be used for the same session. Only one of the two VSAs is expected in the RADIUS Accounting message.</i> Supported on the RADIUS interface in RADIUS Access-Accept messages and MCC generated Accounting messages When this VSA is received in the RADIUS Access-Accept, the PGW/GGSN uses the value received in the VSA to select a data profile for the session. If the value received does not match any configured data profile, the session is terminated. Once the session is set up, the PGW/GGSN populates the TMO-Charging-Rule-Base-Name VSA with the current data profile name in MCC generated RADIUS Accounting-Request messages (Start/Interim/Stop). <i>Note: If both the AF-Rule-Base-Name VSA and the TMO-Charging-Rule-Base-Name VSA are received in the Access-Accept message, the AF-Rule-Base-Name VSA is given priority.</i>

TMO-Transparent-Data

AVP Name	TMO-Transparent-Data
AVP Code	26/29168/2
AVP Data Type	Octet String
Description	Contains Operator-defined transparent data and is used to transport roaming information from the RADIUS server to the PGW via the Access-Accept message. When received in the RADIUS Access-Accept message, the MCC gateway sends the content of this VSA to the OCS via the Gy interface and to the CGF via the Ga interface.

TMO-Virtual-Gi-ID

AVP Name	TMO-Virtual-Gi-ID
AVP Code	26/29168/15
AVP Data Type	32 Bit Integer
Description	Contains the value of the virtual-ID. See Processing the TMO-Virtual-Gi-ID VSA (page 1-56) for more information.

SN-Ecs-Data-Volume

AVP Name	SN-Ecs-Data-Volume
AVP Code	26/8164/176
AVP Data Type	Value contains the following sub-attributes: ■ Rating-Group-Id – Rating Group identifier in a PDP context. – Type: 1 – Length: 4 – Value: Unsigned integer ■ GPRS-Uplink – Uplink octet usage for a PDP context per rating group. – Type: 2 – Length: 4 – Value: Unsigned integer ■ GPRS-Downlink – Downlink octet usage for a PDP context per rating group. – Type: 3 – Length: 4 – Value: Unsigned integer
Description	Contains downlink and uplink octet usage for a PDP context per rating group.

SN1-GTP-Version

AVP Name	SN1-GTP-Version
AVP Code	26/8164/62
AVP Data Type	Integer
Description	Contains the GTP version that the subscriber is using.

SN1-Service-Type

AVP Name	SN1-Service-Type
AVP Code	26/8164/24
AVP Data Type	Integer
Description	Indicates the type of service that the subscriber is accessing.

Sub-Profile-Name

AVP Name	Sub-Profile-Name
AVP Code	26/2352/91
AVP Data Type	String
Description	Received in RADIUS Accounting-Request messages and can be used to select a data profile for GiGW sessions in Subscriber Aware Mode. In Subscriber Aware Mode, subscriber session information is learned from RADIUS Accounting messages via the GiGW session learner (configured with either RADIUS Server or RADIUS Proxy mode). When either the Sub-Profile-Name or the Operator-Charging-Rule-Base-Name VSA is received in the RADIUS Accounting-Request message, the MCC uses the value received in the VSA to select a data profile for the GiGW session and the received value overrides the subscriber analyzer configuration. If an Accounting-Request INTERIM message is received with a data profile name that is different from the one received in the Accounting-Request START message, the MCC switches to the new data profile for that session. <i>Note: The Sub-Profile-Name and Operator-Charging-Rule-Base-Name VSA cannot be used for the same session. Only one of the two VSAs is expected in the RADIUS Accounting message.</i>

3GPP2-Carrier-ID

AVP Name	3GPP2-Carrier-ID
AVP Code	26/5535/142
AVP Data Type	String
Description	Identifies the Mobile Country Code/Mobile Network Code (MCC/MNC) of the visited carrier's PSDN.

3GPP2-GMT-Time-Zone-Offset

AVP Name	3GPP2-GMT-Time-Zone-Offset
AVP Code	26/5535/143
AVP Data Type	Signed Integer
Description	Indicates the time offset (in seconds) at the visited carrier's public switched data network (PSDN) from GMT time (adjusted to reflect standard time and daylight savings time).

